

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\workflows\wf_8d68d336-288\agent-a443ef7f8d7dfe16b.jsonl`  
- SHA-256: `9c24e2bc9534c58257f2f4dd5da3fe4a6d83cd9cf8abadf0957c98d7bad8bd36`  
- Source modified: `2026-07-03T19:52:14+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `wf_8d68d336-288`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T19:48:21.204Z)

Repo (my in-progress optimization branch): C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/fix-worktree

Key files to Read:

- backend/pipeline/detectors/wasb_infer.py ? the inference loop (DataLoader ? detector.run_tensor ? tracker). Contains the streaming SequentialFrameStore, the _PrefetchIter I just added, build_windows, and the tracker re-run loop.
- backend/pipeline/detectors/native_wasb_runner.py ? builds the wasb_infer.py argv on the GPU box (native runner). Has device/stream_video/batch_size/prefetch_batches config.
- backend/config/models.py (WasbIndexConfig) and backend/config/presets.py (cloud-serving preset).
- deploy/cloudrun/Dockerfile ? the WASB conda env is torch 1.11.0+cu113 (py3.8), a SEPARATE env from the app's torch. The detector shells out to it as a subprocess.

The WASB model itself lives at /opt/models/WASB-SBDT (nttcom/WASB-SBDT) ? you can't read it, but reason from how wasb_infer.py calls build_detector(cfg) / detector.run_tensor(imgs, trans) and build_tracker(cfg).

MEASURED BASELINE (first real L4 cloud run, 2026-07-03): a ~7.5-min 60fps badminton clip took Phase-2 (WASB detection) 727.9s (~0.6x realtime); GPU utilization sat <1% the whole time. Hardware: NVIDIA L4 (Ada sm_89, 23GB), 8 vCPU, 32GB. Streaming mode forces DataLoader num_workers=0 (single-process), so CPU batch assembly (cv2 decode + PIL + ToTensor) serializes with GPU inference.

CRITICAL CONSTRAINT ? the D5/D16 byte-parity guarantee: the repo REQUIRES that the rally/window OUTPUT (the candidate windows and trajectory CSV) stay bit-identical / deterministic across runs and across the disk-frames vs streaming paths. This is asserted by a real parity test. An optimization that changes detector NUMERICS (FP16, TF32, cudnn.benchmark algo selection, NVDEC vs CPU decode) risks changing the output and breaking that guarantee. Batch size and prefetch do NOT change per-frame numerics (same forward, just grouped/overlapped) so they are parity-safe. Already implemented on this branch: indexing.wasb.batch_size + prefetch_batches knobs, the cloud-serving preset opts into batch 64 + prefetch 4.

You are a CUDA/PyTorch performance engineer. Investigate ONE optimization lever in depth by READING the actual code. Lever "model-warmup-persist": Investigate two container-lifecycle levers: (1) a warmup forward pass at healthcheck time so the FIRST real batch does not pay lazy CUDA context + cudnn autotune + JIT cost mid-measurement; (2) whether the ephemeral Cloud Run Job model means EVERY job reloads weights + re-JITs ? would a persistent Cloud Run SERVICE (or a torch.jit.trace/save of the model cached in the image) amortize that. Parity-safe framing. Also: is torch.set_grad_enabled already off (torch.no_grad is used) and is model.eval() set?

Read the named files, reason from the real code, and return the structured finding. Be concrete

about code_locations (file:function). Be honest about parity_risk ? the byte-parity guarantee is load-bearing.

2. assistant (2026-07-03T19:48:23.230Z)

I'll investigate the container-lifecycle warmup/persist levers by reading the actual code.

3. user (2026-07-03T19:48:26.088Z)

```
1      """WASB shuttle-trajectory inference on an arbitrary frame sequence (runs in WSL).
2
3      WASB-SBDT only ships *dataset* evaluation; this is a thin wrapper that reuses its
4      own building blocks (ImageDataset transform + detector + online tracker) to run on
5      any folder of extracted frames and emit a trajectory CSV in the exact shape the
6      indexer's `tracknet_runner.parse_trajectory_csv` already consumes:
7
8          Frame,Visibility,X,Y
9
10     It must run INSIDE the WSL `wasb` conda env, from the WASB-SBDT `src/` dir (it
11     imports WASB's `detectors`/`trackers`/`dataloaders`). The Windows-side adapter
12     (`wasb_runner.py`) stages this file + the frames into WSL and invokes it.
13
14     Resumable across reboots
15     -----
16     The GPU detector pass (one forward per sliding window ? ~21k for a 6-min clip) is
17     the slow, expensive part; the CPU tracker pass that turns detections into a
18     trajectory is cheap and *stateful* (it must see every frame in order, so it can't
19     resume mid-stream). We exploit that split:
20
21     * The detector pass writes its per-window raw outputs incrementally to
22       ``<cache>/det_raw.jsonl`` (flushed+fsync'd every ``--flush-every`` windows) and
23       records progress in ``<cache>/manifest.json`` (atomic write). A reboot loses at
24       most ``--flush-every`` windows of GPU work instead of the whole pass.
25     * On restart we replay the cached windows, skip the DataLoader ahead to the first
26       un-cached window, and continue. Once every window is cached, the detector pass
27       is skipped entirely and we just re-run the (cheap, deterministic) tracker over
28       the full ordered cache -> trajectory CSV. So a lost trajectory CSV costs seconds,
29       not the GPU hour.
30
31     The cache is keyed by the output path (``<out>_wasbcache/`` by default), lives next
32     to the output (NOT in %TEMP%), and is invalidated automatically if the frames dir,
33     weights, sport, window size, or frame count change.
34
35     Usage (in WSL, from ~/models/WASB-SBDT/src):
36         python wasb_infer.py --frames_dir /path/frames --weights
37         ../pretrained_weights/wasb_badminton_best.pth.tar \
38         --out /path/traj.csv [--sport badminton] [--limit N] [--cache-dir DIR] [--flush-
39         every 1000] [--fresh]
40
41     """
42
43     import argparse
44     import csv
45     import glob
46     import json
47     import logging
48     import os
49     import os.path as osp
50     from collections import defaultdict
51     from contextlib import contextmanager
52
53     logger = logging.getLogger(__name__)
54
55     CACHE_VERSION = 1
56     _DET_FILE = "det_raw.jsonl"
57     _MANIFEST_FILE = "manifest.json"
```

```

55
56
57     def _build_cfg(weights: str, sport: str, device: str = "cuda"):
58         """Compose the WASB eval config via Hydra, overriding model/weights/device.
59
60         ``device`` is ``cuda`` (default ? the historical WSL behaviour, byte-parity
61         preserved),
62         ``mps`` (Apple) or ``cpu``. Only the single-GPU CUDA path requests
63         ``runner.gpus=[0]``;
64         cpu/mps run with no GPU list so the detector places tensors on ``device``.
65         """
66         from hydra import compose, initialize
67
68         gpus = "[0]" if device == "cuda" else "[]" # single GPU on cuda; none for cpu/mps
69         with initialize(config_path="configs", version_base=None):
70             cfg = compose(
71                 config_name="eval",
72                 overrides=[
73                     f"dataset={sport}",
74                     "model=wasb",
75                     f"detector.model_path={weights}",
76                     f"runner.device={device}",
77                     f"runner.gpus={gpus}", # default config assumes multiple GPUs; pin to
78             this box
79             ],
80             )
81         return cfg
82
83     def _frame_id(path: str) -> int:
84         """Best-effort integer frame id from a frame filename (e.g. frame_000180.png ->
85         180)."""
86         stem = osp.splitext(osp.basename(path))[0]
87         digits = "".join(ch for ch in stem if ch.isdigit())
88         return int(digits) if digits else -1
89
90     # ----- #
91     # Resumable detector cache (pure-Python, no torch/numpy ? unit-testable)
92     # ----- #
93     def _cache_paths(cache_dir: str):
94         return osp.join(cache_dir, _DET_FILE), osp.join(cache_dir, _MANIFEST_FILE)
95
96     def default_cache_dir(out_csv: str) -> str:
97         """Stable cache dir next to the trajectory output (survives reboots; not %TEMP%)."""
98         d = osp.dirname(osp.abspath(out_csv))
99         stem = osp.splitext(osp.basename(out_csv))[0]
100         return osp.join(d, stem + "_wasbcache")
101
102     def _atomic_write_text(path: str, text: str) -> None:
103         """Write then rename so a crash mid-write can't leave a half-written file."""
104         tmp = path + ".tmp"
105         with open(tmp, "w") as f:
106             f.write(text)
107             f.flush()
108             os.fsync(f.fileno())
109         os.replace(tmp, path)
110
111     def _det_plain(det) -> list:
112         """A detector candidate dict {'xy': np.array([x,y]), 'score', 'scale'} -> JSON-able
113         list."""
114         xy = det["xy"]

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115         return [float(xy[0]), float(xy[1]), float(det["score"]), int(det["scale"])]
116
117
118     def _dets_to_tracker(plain_list):
119         """Inverse of _det_plain: rebuild the dicts the tracker expects (xy MUST be a numpy
array,
120         the tracker does vector math / np.linalg.norm on it)."""
121         import numpy as np
122
123         return [
124             {"xy": np.array([p[0], p[1]], dtype=float), "score": p[2], "scale": p[3]}
125             for p in plain_list
126         ]
127
128
129     def write_manifest(cache_dir: str, manifest: dict) -> None:
130         _, mpath = _cache_paths(cache_dir)
131         _atomic_write_text(mpath, json.dumps(manifest, indent=2))
132
133
134     def read_manifest(cache_dir: str):
135         _, mpath = _cache_paths(cache_dir)
136         if not osp.exists(mpath):
137             return None
138         try:
139             with open(mpath) as f:
140                 return json.load(f)
141         except (json.JSONDecodeError, OSError):
142             return None
143
144
145     def manifest_compatible(
146         manifest: dict,
147         *,
148         frames_dir: str,
149         weights: str,
150         sport: str,
151         frames_in: int,
152         frames_total: int,
153         device: str = "cuda",
154     ) -> bool:
155         """A cache is reusable only if the run that produced it matches this run's inputs.
156
157         ``device`` defaults to ``cuda`` so caches written before device-keying (no
158         ``device``
159         field) stay compatible with a CUDA run ? but a cpu/mps run will NOT reuse a cuda
160         cache
161         (detections can differ numerically across devices), and vice-versa.
162
163         ``frames_dir`` is the **already-canonical cache key** the caller stored in the
164         manifest
165         (``run()`` passes its ``cache_key`` here, and writes the same value into the
166         manifest):
167         ``osp.abspath(frames_dir)`` for the frames-on-disk path, or the synthetic
168         ``video:<abspath>``
169         key for the streaming path. We therefore compare it VERBATIM ? do NOT re-``abspath``
170         it here:
171         ``osp.abspath("video:/x.mp4")`` is not recognised as absolute and gets the CWD
172         prepended,
173         so a streaming cache would never match itself and every re-run would recompute
174         (issue #348).
175         """
176         if not manifest or manifest.get("version") != CACHE_VERSION:
177             return False
178         return (

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171         manifest.get("frames_dir") == frames_dir
172         and manifest.get("weights") == weights
173         and manifest.get("sport") == sport
174         and manifest.get("frames_in") == frames_in
175         and manifest.get("frames_total") == frames_total
176         and manifest.get("device", "cuda") == device
177     )
178
179
180 def reset_cache(cache_dir: str) -> None:
181     """Drop any existing cache so the next run starts clean."""
182     det_jsonl, mpath = _cache_paths(cache_dir)
183     for p in (det_jsonl, mpath):
184         if osp.exists(p):
185             os.remove(p)
186
187
188 def load_and_clean_cache(cache_dir: str):
189     """Read the longest *contiguous* (w == line index) valid prefix of det_raw.jsonl,
190     rebuild the per-frame detection accumulation, and rewrite the file to exactly that
191     prefix so a trailing half-written line from a crash can't corrupt future appends.
192
193     Returns (windows_done, det_by_fid) where det_by_fid maps frame_id -> list of
194     [x, y, score, scale] candidates (accumulated across every window the frame is in,
195     in window order ? identical to a non-resumed run).
196     """
197     det_jsonl, _ = _cache_paths(cache_dir)
198     det_by_fid = defaultdict(list)
199     valid: list = []
200     if osp.exists(det_jsonl):
201         with open(det_jsonl, "r") as f:
202             for line in f:
203                 s = line.strip()
204                 if not s:
205                     continue
206                 try:
207                     obj = json.loads(s)
208                 except json.JSONDecodeError:
209                     break # truncated trailing line (crash mid-flush)
210                 if obj.get("w") != len(valid): # non-contiguous -> stop at the gap
211                     break
212                 valid.append(s)
213                 for fid, plain in obj["f"]:
214                     det_by_fid[int(fid)].extend([list(p) for p in plain])
215                 # Guarantee a clean append boundary by rewriting only the good prefix.
216                 _atomic_write_text(det_jsonl, ("\n".join(valid) + "\n") if valid else "")
217     return len(valid), det_by_fid
218
219
220 def _append_windows(fh, objs) -> None:
221     """Append window records as JSONL and durably flush (bounded loss on crash)."""
222     for o in objs:
223         fh.write(json.dumps(o, separators=(",", ":")) + "\n")
224     fh.flush()
225     os.fsync(fh.fileno())
226
227
228 def _atomic_write_trajectory(out_csv: str, rows) -> None:
229     os.makedirs(osp.dirname(osp.abspath(out_csv)), exist_ok=True)
230     tmp = out_csv + ".tmp"
231     with open(tmp, "w", newline="") as f:
232         w = csv.writer(f)
233         w.writerow(["Frame", "Visibility", "X", "Y"])
234         w.writerows(rows)
235         f.flush()

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236         os.fsync(f.fileno())
237         os.replace(tmp, out_csv)
238
239
240     # ----- #
241     # Frame addressing + windowing (pure; no torch/cv2 ? unit-testable)
242     # ----- #
243     _STREAM_FRAME_FMT = "{:08d}.png"
244
245
246     def synthetic_frame_path(index: int) -> str:
247         """Stable, index-encoding frame name used by the streaming path.
248
249         It mirrors the on-disk names ``extract_frames`` writes (``{i:08d}.png``) so the
250         downstream integer-id parser (``_frame_id``) yields the SAME frame id whether the
251         frames came from disk or from the in-memory stream. The streaming image loader
252         inverts this name back to an index to fetch the decoded frame.
253         """
254         return _STREAM_FRAME_FMT.format(int(index))
255
256
257     def build_windows(frames: list, frames_in: int):
258         """Sliding windows of ``frames_in`` consecutive frame *paths* (the unit of GPU work
259         + checkpoint). Pure list math, identical for the disk and streaming paths.
260
261         Returns a list of windows, each a list of ``frames_in`` consecutive paths
262         (``[frames[i:i+frames_in] for i in range(len(frames) - frames_in + 1)]``). The
263         caller wraps each window into the ``{"frames", "annos"}`` sample shape WASB's
264         ``ImageDataset`` expects; keeping that out of here stays torch-free for unit tests.
265         """
266         return [frames[i : i + frames_in] for i in range(len(frames) - frames_in + 1)]
267
268
269     class SequentialFrameStore:
270         """In-memory sliding buffer over a forward-only frame reader, sized for the
271         detector's sliding-window access pattern (peak O(window) frames, not O(video)).
272
273         The detector DataLoader runs with ``shuffle=False`` and (in streaming mode)
274         ``num_workers=0``. ``ImageDataset.__getitem__(i)`` reads the frames of window ``i``
275         in order ? ``i, i+1, ..., i+window-1`` ? and samples are requested ``i = start,
276         start+1, ...``. So the global read sequence dips back by ``window-1`` at each window
277         boundary (``0,1,2, 1,2,3, 2,3,4, ...`` for ``window=3``); the highest index seen so
278         far never decreases. We keep only frames at or above ``max_seen - (window-1)`` (the
279         earliest index any not-yet-finished window can still ask for) and decode each source
280         frame exactly once via the forward-only ``reader(index)``.
281         """
282
283         def __init__(self, reader, total: int, window: int = 1, start_index: int = 0):
284             self._reader = reader
285             self._total = int(total)
286             self._window = max(1, int(window))
287             self._buf: dict = {}
288             # On a resumed run the detector's first needed frame is `start_index`; begin
289             # pulling there so the reader can skip (seek past) the already-cached prefix
290             # instead of decoding 0..start_index-1 just to throw them away.
291             self._next = int(start_index) # next index not yet pulled from the reader
292             self._max_seen = int(start_index) - 1
293             self._floor = int(start_index) # lowest index we still keep (never evict below)
294
295         def get(self, index: int):
296             index = int(index)
297             if index < self._floor:
298                 raise KeyError(
299                     f"frame {index} already evicted (below floor {self._floor}); the access
300                     "

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```

300         f"pattern must stay within a window of the highest frame seen"
301     )
302     if index >= self._total:
303         raise KeyError(f"frame {index} out of range (total {self._total})")
304     # Pull forward from the reader until the requested index is buffered.
305     while self._next <= index:
306         self._buf[self._next] = self._reader(self._next)
307         self._next += 1
308     if index > self._max_seen:
309         self._max_seen = index
310     # Evict frames older than the earliest a still-running window could re-request:
311     # once we've seen index `m`, no future window starts before `m - (window-1)`.
312     new_floor = max(self._floor, self._max_seen - (self._window - 1))
313     for k in range(self._floor, new_floor):
314         self._buf.pop(k, None)
315     self._floor = new_floor
316     return self._buf[index]
317
318
319 def _index_from_synthetic_path(path: str) -> int:
320     """Invert ``synthetic_frame_path``: a frame path -> its integer source index.
321
322     Reuses ``_frame_id`` (digits-only parse) so the index and the cached frame-id stay
323     in lockstep with the on-disk naming scheme.
324     """
325     return _frame_id(path)
326
327
328 def _bgr_to_pil_rgb(frame_bgr):
329     """Convert an in-memory BGR uint8 frame (as ``cv2.VideoCapture`` yields) to the
330     EXACT
331     PIL RGB image WASB's disk path produces.
332
333     The disk path is ``frame_bgr -> cv2.imwrite(PNG) ->
334     Image.open(PNG).convert('RGB')``;
335     because PNG is lossless that round-trip equals ``cv2.cvtColor(frame_bgr,
336     COLOR_BGR2RGB)``
337     bit-for-bit (verified empirically, max|diff|=0). Returning that here makes the
338     streamed
339     frame indistinguishable from the on-disk one to everything downstream.
340     """
341     import cv2
342     from PIL import Image
343
344     return Image.fromarray(cv2.cvtColor(frame_bgr, cv2.COLOR_BGR2RGB))
345
346
347 def _make_streamed_read_image(store, real_read_image):
348     """Build the ``read_image`` replacement used during a streaming detector pass.
349
350     A synthetic frame path -> the in-memory decoded frame for its index (as a PIL RGB
351     image,
352     matching ``read_image``'s real return type); any path that doesn't parse to an index
353     falls
354     through to ``real_read_image``. Pure ? imports no WASB module ? so it is unit-
355     testable
356     without the WSL-only ``dataloaders`` package.
357     """
358
359     def _read_image(path, *args, **kwargs):
360         idx = _index_from_synthetic_path(path)
361         if idx < 0:
362             return real_read_image(path, *args, **kwargs)
363         return _bgr_to_pil_rgb(store.get(idx))
364

```

```

358         return _read_image
359
360
361     @contextmanager
362     def _streaming_read_image_patch(
363         frame_loader, total: int, window: int = 1, start_index: int = 0
364     ):
365         """Scope a shim over WASB's ``read_image`` so ``ImageDataset`` is fed in-memory
366         decoded frames during the detector pass, byte-for-byte as it would over real PNGs.
367
368         WASB's ``ImageDataset.__getitem__`` reads each frame via ``read_image(path)`` ? NOT
369         ``cv2.imread``. ``read_image`` (``utils.utils``) does
370         ``Image.open(path).convert('RGB')``,
371         returning a PIL **RGB** image, after an ``osp.exists(path)`` guard. We therefore
372         feed it a
373         PIL RGB image built from the streamed BGR frame (see ``_bgr_to_pil_rgb``), which is
374         byte-identical to the disk round-trip, and the shim never touches the filesystem so
375         the
376         ``osp.exists`` guard is sidestepped for synthetic stream paths.
377
378         We patch ``dataloaders.dataset_loader.read_image`` ? the binding the consumer
379         actually
380         calls (``dataset_loader`` does ``from utils import read_image``, so the name lives
381         in that
382         module's namespace; patching ``utils.read_image`` would NOT rebind the already-
383         imported
384         reference).
385
386         ``frame_loader(index) -> BGR uint8 ndarray`` is a forward-only sequential decoder
387         wrapped
388         in a ``SequentialFrameStore`` (sliding buffer sized to ``window`` = ``frames_in``);
389         on a
390         resume we start at ``start_index`` so the decoder seeks past already-cached windows.
391         The
392         original reader is restored on exit.
393         """
394         from dataloaders import dataset_loader as _dl
395
396         store = SequentialFrameStore(
397             frame_loader, total, window=window, start_index=start_index
398         )
399         real_read_image = _dl.read_image
400         # Rebind read_image in the consuming module for the duration of the detector pass;
401         # restore on exit. (Plain assignment, not setattr-with-constant, to stay ruff
402         B010-clean.)
403         _dl.read_image = _make_streamed_read_image(store, real_read_image) # type:
404         ignore[assignment]
405         try:
406             yield store
407         finally:
408             _dl.read_image = real_read_image # type: ignore[assignment]
409
410     # ----- #
411     # Inference
412     # ----- #
413     class _PrefetchIter:
414         """Bounded producer-thread wrapper over a batch iterator (the num_workers=0
415         DataLoader).
416
417         In streaming mode the DataLoader must stay single-process (the in-memory frame store
418         +
419         the cv2.imread shim live in this process and cannot be pickled to workers), so the
420         CPU-side batch assembly (decode + PIL + ToTensor) SERIALIZES with GPU inference ?

```

```

409     measured on the first real L4 serving run (2026-07-03) this left the GPU <1%
utilized.
410     A single producer THREAD fixes the overlap without multiprocessing: cv2/PIL/numpy
411     release the GIL for the heavy parts, the store's forward-only access pattern is
412     preserved (one consumer thread iterating sequentially), and the bounded queue caps
413     memory at ``depth`` batches. A producer exception is re-raised in the consumer."""
414
415     _END = object()
416
417     def __init__(self, src, depth: int = 2):
418         import queue
419         import threading
420
421         self._q: "queue.Queue" = queue.Queue(maxsize=max(1, int(depth)))
422         self._exc: "BaseException | None" = None
423
424         def _fill() -> None:
425             try:
426                 for item in src:
427                     self._q.put(item)
428             except BaseException as e: # noqa: BLE001 - surfaced in __next__
429                 self._exc = e
430             finally:
431                 self._q.put(self._END)
432
433         self._t = threading.Thread(target=_fill, name="wasb-prefetch", daemon=True)
434         self._t.start()
435
436     def __iter__(self):
437         return self
438
439     def __next__(self):
440         item = self._q.get()
441         if item is self._END:
442             if self._exc is not None:
443                 raise self._exc
444             raise StopIteration
445         return item
446
447
448     def run(
449         frames,
450         weights: str,
451         out_csv: str,
452         sport: str = "badminton",
453         limit: int = 0,
454         batch_size: int = 8,
455         num_workers: int = 4,
456         log_every_batches: int = 50,
457         cache_dir: "str | None" = None,
458         flush_every: int = 1000,
459         fresh: bool = False,
460         frame_loader=None,
461         source_key: "str | None" = None,
462         device: str = "cuda",
463         prefetch_batches: int = 0,
464     ) -> str:
465         """Run WASB detection+tracking over an ordered frame source -> trajectory CSV.
466
467         Two equivalent ways to supply pixels (output is bit-identical between them):
468
469         * **frames-on-disk** (default): ``frames`` is a directory path string; frames are
470         globbed (``*.png``/``*.jpg``) and ``ImageDataset`` reads each file via
471         ``cv2.imread``.
472
473         * **streaming** (fast path): ``frames`` is a pre-built ordered list of (synthetic)

```

```

472     frame paths and ``frame_loader(index) -> BGR uint8 ndarray`` supplies the decoded
473     pixels in-memory. We scope a shim over WASB's ``read_image`` (the actual per-frame
474     reader, which returns a PIL RGB image) over the DataLoader pass so every other
line of
475     ``ImageDataset.__getitem__`` runs UNCHANGED ? the tensor the detector sees is
identical
476     to the disk round-trip (``cv2.imwrite`` PNG -> ``Image.open().convert('RGB')`` ==
477     ``cvtColor(bgr, BGR2RGB)``), verified byte-identical). Streaming forces
``num_workers=0``
478     (the in-process shim + buffer can't cross worker processes).
479
480     ``source_key`` overrides the cache-keying identity (defaults to the abspath of the
481     frames dir); the streaming path passes a stable ``video:<abspath>`` key so a resume
482     after reboot still matches.
483     """
484     import time
485
486     import torch
487     from dataloaders import build_img_transforms, build_seq_transforms
488     from dataloaders.dataset_loader import ImageDataset
489     from torch.utils.data import DataLoader
490
491     # NB: this is WASB's OWN local `trackers` module (from its repo, on sys.path inside
the
492     # `wasb` conda env), NOT the Roboflow `trackers` package. The `type: ignore`
suppresses
493     # the mypy false positive where the main env resolves `trackers` to the Roboflow
package
494     # (which has no `build_tracker`). There is NO runtime collision ? this runs in a
separate
495     # subprocess/conda env. Do NOT "clean up" the ignore without removing the Roboflow
dep.
496     from trackers import build_tracker # type: ignore[attr-defined]
497     from utils import Center
498
499     from detectors import build_detector
500
501     streaming = frame_loader is not None
502
503     cfg = _build_cfg(weights, sport, device)
504     frames_in = int(cfg["model"]["frames_in"])
505     input_wh = (int(cfg["model"]["inp_width"]), int(cfg["model"]["inp_height"]))
506     output_wh = (int(cfg["model"]["out_width"]), int(cfg["model"]["out_height"]))
507
508     if streaming:
509         # `frames` is already the ordered list of (synthetic) frame paths.
510         if not isinstance(frames, (list, tuple)):
511             raise SystemExit(
512                 "streaming mode requires `frames` to be a list of frame paths"
513             )
514         frames = list(frames)
515         frames_dir = source_key or "stream"
516     else:
517         frames_dir = frames
518         frames = sorted(
519             glob.glob(osp.join(frames_dir, "*.png"))
520             + glob.glob(osp.join(frames_dir, "*.jpg"))
521         )
522     if limit:
523         frames = frames[:limit]
524     if len(frames) < frames_in:
525         where = frames_dir if not streaming else "video stream"
526         raise SystemExit(f"need >= {frames_in} frames, found {len(frames)} in {where}")
527
528     # Sliding windows of `frames_in` consecutive frames (the unit of GPU work +

```

```

checkpoint).
529     samples = []
530     for window in build_windows(frames, frames_in):
531         annos = [
532             {"frame_path": p, "center": Center(is_visible=False, x=-1.0, y=-1.0)}
533             for p in window
534         ]
535     samples.append({"frames": window, "annos": annos})
536 windows_total = len(samples)
537
538 # ---- cache setup / resume decision ----- #
539 if cache_dir is None:
540     cache_dir = default_cache_dir(out_csv)
541     os.makedirs(cache_dir, exist_ok=True)
542     det_jsonl, _ = _cache_paths(cache_dir)
543     cache_key = source_key if source_key is not None else osp.abspath(frames_dir)
544     manifest = None if fresh else read_manifest(cache_dir)
545     resume = manifest_compatible(
546         manifest or {},
547         frames_dir=cache_key,
548         weights=weights,
549         sport=sport,
550         frames_in=frames_in,
551         frames_total=len(frames),
552         device=device,
553     )
554     if resume:
555         windows_done, det_by_fid = load_and_clean_cache(cache_dir)
556         logger.info(
557             f"resuming from cache: {windows_done}/{windows_total} windows already
558 detected"
559         )
560     else:
561         if manifest is not None:
562             logger.info("cache incompatible with this run -> starting fresh")
563             reset_cache(cache_dir)
564             windows_done, det_by_fid = 0, defaultdict(list)
565
566     base_manifest = {
567         "version": CACHE_VERSION,
568         "frames_dir": cache_key,
569         "weights": weights,
570         "sport": sport,
571         "frames_in": frames_in,
572         "frames_total": len(frames),
573         "device": device,
574         "windows_total": windows_total,
575         "windows_done": windows_done,
576     }
577     write_manifest(cache_dir, base_manifest)
578
579 # ---- detector pass over the un-cached windows ----- #
580 remaining = samples[windows_done:]
581 if remaining:
582     _, transform_test = build_img_transforms(cfg)
583     try:
584         _, seq_transform_test = build_seq_transforms(cfg)
585     except Exception:
586         seq_transform_test = None
587     ds = ImageDataset(
588         cfg,
589         remaining,
590         input_wh,
591         output_wh,
592         transform=transform_test,

```

```

592         seq_transform=seq_transform_test,
593         is_train=False,
594     )
595     # Streaming forces single-process loading: the in-memory frame store + the
596     # cv2.imread shim live in THIS process and can't be pickled to DataLoader
workers.
597     loader_workers = 0 if streaming else num_workers
598     loader = DataLoader(
599         ds, batch_size=batch_size, shuffle=False, num_workers=loader_workers
600     )
601     detector = build_detector(cfg)
602
603     from contextlib import ExitStack
604
605     t0 = time.time()
606     done = windows_done
607     win_idx = windows_done
608     log_every = max(1, log_every_batches)
609     flush_n = max(1, flush_every)
610     pending = []
611     logger.info(
612         f"detecting on {len(remaining)} remaining windows "
613         f"(total {windows_total}, batch={batch_size}, workers={loader_workers}, "
614         f"source={'video-stream' if streaming else 'frames-dir'})..."
615     )
616     det_f = open(det_jsonl, "a")
617     try:
618         with ExitStack() as stack:
619             stack.enter_context(torch.no_grad())
620             # In streaming mode, route ImageDataset's per-frame `cv2.imread(path)`
to the
621             # in-memory decoded frame for that synthetic path; every other line of
622             # __getitem__ runs unchanged so the tensor matches the PNG round-trip
exactly.
623             # The detector's first needed frame is `windows_done` (window i starts
at
624             # frame i), so prime the store there for a resume.
625             if streaming:
626                 stack.enter_context(
627                     _streaming_read_image_patch(
628                         frame_loader,
629                         len(frames),
630                         window=frames_in,
631                         start_index=windows_done,
632                     )
633                 )
634             # Overlap CPU batch assembly with GPU inference (streaming/workers=0
only ?
635             # a multi-worker DataLoader already overlaps via its worker processes).
636             batches = (
637                 _PrefetchIter(loader, depth=prefetch_batches)
638                 if prefetch_batches and loader_workers == 0
639                 else loader
640             )
641             for bi, batch in enumerate(batches):
642                 imgs, _hms, trans = batch[0], batch[1], batch[2]
643                 img_paths = [
644                     list(t) for t in batch[-1]
645                 ] # [frame_in][batch] -> path
646                 batch_results, _ = detector.run_tensor(imgs, trans)
647                 nb = imgs.shape[0]
648                 for ib in range(nb):
649                     frames_payload = []
650                     for ie in sorted(batch_results[ib].keys()):
651                         p = img_paths[ie][ib]

```

```

652         fid = _frame_id(p)
653         plain = [_det_plain(d) for d in batch_results[ib][ie]]
654         frames_payload.append([fid, plain])
655         det_by_fid[fid].extend(plain)
656         pending.append({"w": win_idx, "f": frames_payload})
657         win_idx += 1
658     done += nb
659     if len(pending) >= flush_n or done >= windows_total:
660         _append_windows(det_f, pending)
661         pending = []
662         base_manifest["windows_done"] = done
663         write_manifest(cache_dir, base_manifest)
664     if (bi + 1) % log_every == 0 or done >= windows_total:
665         el = time.time() - t0
666         new_done = done - windows_done
667         rate = new_done / el if el > 0 else 0.0
668         eta = (windows_total - done) / rate if rate > 0 else 0.0
669         logger.info(
670             f"{done}/{windows_total} "
671             f"({100 * done // max(1, windows_total)}%) | {rate:.1f}
win/s | "
672             f"elapsed {el:.0f}s | ETA {eta:.0f}s"
673         )
674     if pending:
675         _append_windows(det_f, pending)
676         base_manifest["windows_done"] = done
677         write_manifest(cache_dir, base_manifest)
678     finally:
679         det_f.close()
680     logger.info(f"detection done in {time.time() - t0:.0f}s; running tracker...")
681 else:
682     logger.info("detector cache already complete -> tracker-only re-run")
683
684 # ---- tracker re-run over the full ordered cache (cheap, deterministic) ---- #
685 tracker = build_tracker(cfg)
686 tracker.refresh()
687 rows = []
688 for fid in sorted(det_by_fid.keys()):
689     r = tracker.update(_dets_to_tracker(det_by_fid[fid]))
690     rows.append((fid, 1 if r.get("visi") else 0, r.get("x", -1), r.get("y", -1)))
691
692 _atomic_write_trajectory(out_csv, rows)
693 visible = sum(1 for _, v, *_ in rows if v)
694 logger.info(f"wrote {len(rows)} rows -> {out_csv}")
695 logger.info(f"visible-shuttle frames: {visible}/{len(rows)}")
696 return out_csv
697
698
699 def extract_frames(video_path: str, frames_dir: str) -> str:
700     """Decode a video to PNG frames named by source frame index (00000000.png ...).
701
702     Idempotent: if the dir already holds the expected number of frames (matching the
703     container's reported frame count) we skip re-decoding ? re-extraction after a
704     reboot is wasteful. NOTE: cv2 PNG extraction is slow; for big clips prefer
705     extracting with ffmpeg on the host and passing --frames_dir.
706     """
707     import cv2
708
709     os.makedirs(frames_dir, exist_ok=True)
710     cap = cv2.VideoCapture(video_path)
711     if not cap.isOpened():
712         raise SystemExit(f"cannot open video: {video_path}")
713     expected = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
714     existing = len(glob.glob(osp.join(frames_dir, "*.png")))
715     if expected > 0 and existing >= expected:

```

```

716         cap.release()
717         logger.info(f"reusing {existing} already-extracted frames -> {frames_dir}")
718         return frames_dir
719     i = 0
720     while True:
721         ok, frame = cap.read()
722         if not ok:
723             break
724         cv2.imwrite(osp.join(frames_dir, f"{i:08d}.png"), frame)
725         i += 1
726     cap.release()
727     logger.info(f"extracted {i} frames -> {frames_dir}")
728     return frames_dir
729
730
731 def _probe_frame_count(video_path: str) -> int:
732     """Container-reported frame count via cv2 (0 if unknown). Used to size the
733     stream."""
734     import cv2
735     cap = cv2.VideoCapture(video_path)
736     if not cap.isOpened():
737         raise SystemExit(f"cannot open video: {video_path}")
738     n = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
739     cap.release()
740     return n
741
742
743 def make_video_frame_loader(video_path: str):
744     """Return ``(frame_loader, count_hint, release)`` for output-preserving streaming.
745
746     ``frame_loader(index) -> BGR uint8 ndarray`` decodes the video forward-only with one
747     long-lived ``cv2.VideoCapture``. It MUST be called with non-decreasing indices (the
748     detector's sliding-window order); it reads sequentially and only uses a frame-peek
749     to
750     skip forward when a *resume* starts past the current position. Each ``cap.read()``
751     returns exactly the BGR uint8 array that ``cv2.imwrite``/``cv2.imread`` of a PNG
752     would
753     yield (PNG is lossless), so the detector sees identical pixels to the disk path.
754
755     Returns the count hint from the container so the caller can build the frame list;
756     ``release`` closes the capture.
757     """
758     import cv2
759     cap = cv2.VideoCapture(video_path)
760     if not cap.isOpened():
761         raise SystemExit(f"cannot open video: {video_path}")
762     count_hint = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
763     state = {"pos": 0} # next index cap.read() will return
764
765     def frame_loader(index: int):
766         index = int(index)
767         if index < state["pos"]:
768             raise RuntimeError(
769                 f"streaming decode is forward-only; got index {index} after
770                 {state['pos']}")
771         if index > state["pos"]:
772             # Forward skip only happens when resuming past cached windows. Seek once.
773             cap.set(cv2.CAP_PROP_POS_FRAMES, index)
774             state["pos"] = index
775         ok, frame = cap.read()
776         if not ok:
777             raise RuntimeError(f"failed to decode frame {index} from {video_path}")

```

```

777         state["pos"] += 1
778         return frame
779
780     def release():
781         cap.release()
782
783     return frame_loader, count_hint, release
784
785
786 def run_video_streaming(
787     video_path: str,
788     weights: str,
789     out_csv: str,
790     sport: str = "badminton",
791     limit: int = 0,
792     batch_size: int = 8,
793     log_every_batches: int = 50,
794     cache_dir: "str | None" = None,
795     flush_every: int = 1000,
796     fresh: bool = False,
797     device: str = "cuda",
798     prefetch_batches: int = 0,
799 ) -> str:
800     """Fast path: decode ``video_path`` on the fly and feed frames straight to the
801     detector, WITHOUT extracting every frame to a PNG first (the disk round-trip that
802     dominates a long clip). Output is bit-identical to the frames-on-disk path; see
803     ``run``'s docstring for why.
804     """
805     frame_loader, count_hint, release = make_video_frame_loader(video_path)
806     try:
807         if count_hint <= 0:
808             raise SystemExit(
809                 f"could not determine frame count for {video_path}; cannot stream "
810                 f"(fall back to frame extraction with --frames_dir)"
811             )
812         frame_list = [synthetic_frame_path(i) for i in range(count_hint)]
813         source_key = "video:" + osp.abspath(video_path)
814         return run(
815             frame_list,
816             weights,
817             out_csv,
818             sport,
819             limit,
820             batch_size=batch_size,
821             num_workers=0,
822             log_every_batches=log_every_batches,
823             cache_dir=cache_dir,
824             flush_every=flush_every,
825             fresh=fresh,
826             frame_loader=frame_loader,
827             source_key=source_key,
828             device=device,
829             prefetch_batches=prefetch_batches,
830         )
831     finally:
832         release()
833
834
835 def main():
836     ap = argparse.ArgumentParser()
837     ap.add_argument("--frames_dir", help="folder of pre-extracted frames")
838     ap.add_argument(
839         "--video", help="video file; frames are extracted internally (in WSL)"
840     )
841     ap.add_argument(

```

```

842         "--frames_out_dir", help="where to extract frames when --video is used"
843     )
844     ap.add_argument(
845         "--stream-video",
846         action="store_true",
847         help="FAST PATH: decode --video on the fly and feed frames straight to "
848         "the detector (no PNG extraction). Output-equivalent to the disk path.",
849     )
850     ap.add_argument("--weights", required=True)
851     ap.add_argument("--out", required=True)
852     ap.add_argument("--sport", default="badminton")
853     ap.add_argument(
854         "--device",
855         default="cuda",
856         choices=["cuda", "mps", "cpu"],
857         help="torch device for the detector (default cuda = the WSL parity path)",
858     )
859     ap.add_argument(
860         "--limit", type=int, default=0, help="cap #frames (for quick tests)"
861     )
862     ap.add_argument("--batch-size", type=int, default=8)
863     ap.add_argument("--num-workers", type=int, default=4)
864     ap.add_argument(
865         "--prefetch-batches",
866         type=int,
867         default=0,
868         help="streaming mode: overlap CPU batch assembly with GPU inference via a "
869         "bounded producer thread of this queue depth (0 = off, the parity default)",
870     )
871     ap.add_argument(
872         "--log-every-batches", type=int, default=50, help="progress log cadence"
873     )
874     ap.add_argument(
875         "--cache-dir",
876         default=None,
877         help="resumable detector cache dir (default: <out>_wasbcache next to --out)",
878     )
879     ap.add_argument(
880         "--flush-every",
881         type=int,
882         default=1000,
883         help="fsync the detector cache every N windows (bounds reboot loss)",
884     )
885     ap.add_argument(
886         "--fresh", action="store_true", help="ignore any existing cache and start over"
887     )
888     args = ap.parse_args()
889
890     if args.stream_video:
891         if not args.video:
892             raise SystemExit("--stream-video requires --video")
893         run_video_streaming(
894             args.video,
895             args.weights,
896             args.out,
897             args.sport,
898             args.limit,
899             batch_size=args.batch_size,
900             log_every_batches=args.log_every_batches,
901             cache_dir=args.cache_dir,
902             flush_every=args.flush_every,
903             fresh=args.fresh,
904             device=args.device,
905             prefetch_batches=args.prefetch_batches,
906         )

```

```

907         return
908
909     frames_dir = args.frames_dir
910     if args.video:
911         frames_dir = extract_frames(
912             args.video, args.frames_out_dir or (args.video + "_frames")
913         )
914     if not frames_dir:
915         raise SystemExit("provide --frames_dir or --video")
916     run(
917         frames_dir,
918         args.weights,
919         args.out,
920         args.sport,
921         args.limit,
922         batch_size=args.batch_size,
923         num_workers=args.num_workers,
924         log_every_batches=args.log_every_batches,
925         cache_dir=args.cache_dir,
926         flush_every=args.flush_every,
927         fresh=args.fresh,
928         device=args.device,
929     )
930
931
932 if __name__ == "__main__":
933     main()
934

```

4. user (2026-07-03T19:48:26.731Z)

```

1      """Linux-native WASB shuttle-trajectory runner (no WSL).
2
3      The de-risk goal the ``DetectorRunner`` ABC was built for (see ``base.py`` ?
4      "Optional Linux-native path"; ``docs/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md``
M3.5).
5      It is the sibling of :class:`~backend.pipeline.detectors.wasb_runner.WasbRunner` for a
6      Linux GPU box (GCP / a SaaS worker): it runs the same verified ``wasb_infer.py``
core
7      that the WSL adapter runs, so the trajectory CSV is identical (to a sub-pixel float
tolerance;
8      see the acceptance gate below) ? but WITHOUT the WSL transport:
9
10     * NO ``WslCommandMixin`` (no ``wsl -d <distro> bash -c``), NO ``conda activate`` (the
11     ``wasb`` env's python is invoked by path), NO ``/mnt/c`` path translation
12     (``to_wsl_mnt_path``). Paths are native; the video is already local on the box.
13     * ``device`` is ``cuda`` (default ? the WSL parity path), ``mps``, ``cpu``, or
``auto``
14     (M4: cuda-if-available-else-cpu ? the CPU fallback for a no-GPU box; explicit
``cuda`` stays
15     strict); threaded into ``wasb_infer.py --device`` (and so the Hydra
``runner.device`` override).
16     * weights / repo / python / scratch come from the environment first
17     (``WASB_WEIGHTS`` / ``WASB_REPO_DIR`` / ``WASB_PYTHON`` / ``WASB_SCRATCH``), then
18     ``indexing.wasb.*`` config, then a default ? so a box needs no config edits.
19
20     It emits the identical ``Frame,Visibility,X,Y`` CSV (same ``{stem}__{vid12}_wasb.csv``
21     name as the WSL runner) and reuses the model-agnostic ``parse_trajectory_csv`` /
22     ``trajectory_to_action_windows`` from ``tracknet_runner``, so the trajectory hybrids and
23     the windowing brain are unchanged. Selected via the existing ``detector_impl`` seam
24     (``trajectory_hybrid.resolve_runner`` ? ``_build_native_runner``);
``detector_impl="native"``.
25
26     Acceptance gate (parity, NOT strict byte-equality): the SAME clip through the WSL
runner

```

```

27     (Windows) and this runner (Linux, ``device=cuda``) yields the same trajectory CSV + the
same
28     candidate windows **within a small float tolerance (~1e-3 px)**. cuDNN is non-
deterministic
29     ACROSS GPUs, so a sub-pixel diff is expected on different hardware ? but it is byte-
exact
30     run-to-run on a FIXED GPU. ? Validated 2026-06-20 on an RTX 2070 (via WSL): native
streaming
31     vs the cached WSL-disk CSV matched **1194/1195** frames (the lone diff a ~5e-5 px edge
32     artifact), and two native runs were byte-identical (deterministic). This is a hardware
check
33     (a real GPU + the ``wasb`` env), so the offline, subprocess-mocked suite asserts the
contract,
34     not the numbers.
35     """
36
37     import logging
38     import os
39     import shutil
40     import subprocess
41     from dataclasses import dataclass
42     from typing import List, Optional, Tuple
43
44     from backend.config.models import IndexingConfig
45     from backend.pipeline.detectors.base import DetectorRunner
46     from backend.pipeline.detectors.device import AUTO, VALID_DEVICES, DeviceContext
47
48     # Reuse the model-agnostic helpers ? trajectory shape + windowing are identical to
WSL/TrackNet.
49     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
50
51     logger = logging.getLogger(__name__)
52
53     # Path to the inference core we copy into the WASB repo's src/ (so it can import WASB's
54     # detectors/trackers/dataloaders + find configs/), exactly as the WSL adapter does.
55     _INFER_PY = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
56
57     # Single torch/cuda probe code, used by BOTH healthcheck and the lazy device-resolution
probe
58     # so the "cuda True/False" string they parse can never drift apart.
59     _CUDA_PROBE_CODE = (
60         "import torch; print('torch', torch.__version__, 'cuda', torch.cuda.is_available())"
61     )
62
63
64     @dataclass
65     class NativeWasbConfig:
66         """Resolved settings for a Linux-native WASB run.
67
68         Built via :meth:`from_indexing_cfg` with **environment overrides** taking precedence
69         over ``indexing.wasb.*`` config (the box sets env; no config edits needed).
``device``
70         defaults to ``cuda`` (the WSL parity path); ``stream_video`` defaults to True (the
perf win).
71         """
72
73         repo_dir: str = "~/models/WASB-SBDT" # native clone of nttcom/WASB-SBDT
74         python_bin: str = (
75             "python" # the `wasb` conda env python (invoked by path, no activate)
76         )
77         weights_path: str = "" # REQUIRED (env WASB_WEIGHTS or indexing.wasb.weights_path)
78         sport: str = "badminton"
79         device: str = "cuda" # auto | cuda | mps | cpu (auto = cuda-if-available-else-cpu)
80         scratch_dir: str = "" # frame-extraction scratch (env); "" = next to the output dir
81         timeout_sec: int = 1800 # 30 min; a hung GPU call is killed (0 = no timeout)

```

```

82     keep_frames: bool = (
83         False # retain the decoded frame cache after success (debug only)
84     )
85     # NATIVE DEFAULT = STREAMING: skip the cv2 PNG extraction that dominates a long
clip's
86     # wall-clock (measured ~50s on a 20s clip) ? bit-identical to disk (verified on a
real GPU
87     # 2026-06-20). The GPU box reads from a local NVMe so streaming is the right default
here.
88     stream_video: bool = True
89     # GPU-feed tuning (indexing.wasb.batch_size / prefetch_batches). 0 = omit the flag ?
90     # wasb_infer.py's parity defaults (batch 8, no prefetch). The cloud-serving preset
opts in.
91     batch_size: int = 0
92     prefetch_batches: int = 0
93
94     @classmethod
95     def from_indexing_cfg(cls, idx_cfg: "IndexingConfig") -> "NativeWasbConfig":
96         if isinstance(idx_cfg, dict):
97             idx_cfg = IndexingConfig(**idx_cfg)
98         w = idx_cfg.wasb
99         env = os.environ.get
100         # stream_video is tri-state in config: None/absent => native default (stream);
an
101         # explicit True/False still wins (e.g. False for a container cv2 can't seek).
102         sv = w.stream_video
103         return cls(
104             repo_dir=env("WASB_REPO_DIR") or w.repo_dir,
105             python_bin=env("WASB_PYTHON") or w.python_bin,
106             weights_path=env("WASB_WEIGHTS") or w.weights_path or "",
107             sport=w.sport,
108             device=env("WASB_DEVICE") or w.device,
109             scratch_dir=env("WASB_SCRATCH")
110             or env("RALLY_SCRATCH")
111             or env("TMPDIR")
112             or "",
113             timeout_sec=w.timeout_sec,
114             keep_frames=w.keep_frames,
115             stream_video=(True if sv is None else bool(sv)),
116             batch_size=int(w.batch_size or 0),
117             prefetch_batches=int(w.prefetch_batches or 0),
118         )
119
120
121     class NativeWasbRunner(DetectorRunner):
122         """WASB runner that executes natively on a Linux GPU box ? no WSL. See module
docstring."""
123
124         def __init__(self, cfg: NativeWasbConfig):
125             self.cfg = cfg
126             # Cached CUDA-availability probe (used to resolve device="auto"). Set by
healthcheck;
127             # lazily probed by run_predict if healthcheck was skipped. None = not yet
probed.
128             self._cuda_available: Optional[bool] = None
129
130             # --- low-level native exec (the single place tests patch)
-----
131         def _run(
132             self, argv: List[str], cwd: Optional[str] = None
133         ) -> subprocess.CompletedProcess:
134             logger.debug("native exec: %s (cwd=%s)", " ".join(argv), cwd)
135             return subprocess.run(
136                 argv,
137                 cwd=cwd,

```

```

138         capture_output=True,
139         text=True,
140         timeout=(self.cfg.timeout_sec or None),
141     )
142
143     # --- device resolution (M4 DeviceContext: device="auto" ? cuda-if-available-else-
cpu) -----
144     def _probe_cuda(self) -> bool:
145         """Probe ``torch.cuda.is_available()`` in the wasb env's python (a subprocess).
Any failure
146         (missing python / timeout / import error) is treated as 'no CUDA'."""
147         try:
148             res = self._run([self.cfg.python_bin, "-c", _CUDA_PROBE_CODE])
149         except (FileNotFoundError, subprocess.TimeoutExpired):
150             return False
151         return res.returncode == 0 and "cuda True" in (res.stdout or "")
152
153     def _cuda_is_available(self) -> bool:
154         """CUDA availability, cached across healthcheck + run_predict so we probe at
most once."""
155         if self._cuda_available is None:
156             self._cuda_available = self._probe_cuda()
157         return self._cuda_available
158
159     def _effective_device(self) -> str:
160         """The torch device to actually pass to ``wasb_infer.py``. Only ``auto`` needs
the CUDA
161         probe; an explicit cuda/mps/cpu resolves to itself (so ``device='cuda'`` is
unchanged)."""
162         if self.cfg.device == AUTO:
163             return DeviceContext(AUTO, self._cuda_is_available()).effective
164         return self.cfg.device
165
166     def _repo_src(self) -> str:
167         return os.path.join(os.path.expanduser(self.cfg.repo_dir), "src")
168
169     def _copy_infer_script(self) -> bool:
170         """Copy wasb_infer.py into the WASB repo src/ so it imports WASB modules + finds
configs/."""
171         repo_src = self._repo_src()
172         try:
173             os.makedirs(repo_src, exist_ok=True)
174             shutil.copy(_INFER_PY, os.path.join(repo_src, "wasb_infer.py"))
175             return True
176         except OSError as e:
177             logger.error("Failed to copy wasb_infer.py into %s: %s", repo_src, e)
178             return False
179
180     def _rm_rf(self, path: Optional[str]) -> None:
181         """Best-effort recursive delete of a native path (post-success cleanup; never
raises)."""
182         if path:
183             shutil.rmtree(path, ignore_errors=True)
184
185     def healthcheck(self) -> Tuple[bool, str]:
186         """Verify weights + repo present and the `wasb` python imports torch (and sees a
GPU on cuda)."""
187         if not self.cfg.weights_path:
188             return False, (
189                 "WASB weights path is empty ? set WASB_WEIGHTS (or
indexing.wasb.weights_path)."
190             )
191         weights = os.path.expanduser(self.cfg.weights_path)
192         if not os.path.exists(weights):
193             return False, f"WASB weights not found at {weights}"

```

```

194         repo_src = self._repo_src()
195         if not os.path.isdir(repo_src):
196             return False, (
197                 f"WASB-SBDT repo src/ not found at {repo_src} ? clone nttcom/WASB-SBDT "
198                 f"(set WASB_REPO_DIR / indexing.wasb.repo_dir).")
199         )
200         # Fail fast on a typo'd device (the device-policy seam) ? a clear error here
beats a cryptic
201         # argparse rc=2 from wasb_infer.py at run time.
202         if self.cfg.device not in VALID_DEVICES:
203             return False, (
204                 f"unknown device {self.cfg.device!r} ? valid: {'',
205                 '.join(VALID_DEVICES)}')."
206             )
207         try:
208             res = self._run([self.cfg.python_bin, "-c", _CUDA_PROBE_CODE])
209         except FileNotFoundError:
210             return (
211                 False,
212                 f"python binary not found: {self.cfg.python_bin!r} (set WASB_PYTHON).",
213             )
214         except subprocess.TimeoutExpired:
215             return False, "native torch healthcheck timed out."
216         out = (res.stdout or "").strip()
217         if res.returncode != 0:
218             return (
219                 False,
220                 f"`{self.cfg.python_bin}` torch import failed: {(res.stderr or
221                 out).strip()}",
222             )
223         # M4: resolve the device. Cache the probe so run_predict reuses it (no second
subprocess).
224         self._cuda_available = "cuda True" in out
225         ctx = DeviceContext(self.cfg.device, self._cuda_available)
226         if ctx.cuda_required_but_missing:
227             # Unchanged pre-M4 behaviour for an EXPLICIT cuda request: hard-fail, never
a silent
228             # slow-CPU downgrade. Set indexing.wasb.device=auto for a CPU fallback.
229             return False, f"device=cuda but torch.cuda.is_available() is False ({out})"
230         if ctx.fell_back:
231             logger.warning(
232                 "device=auto: no CUDA GPU detected (%s) ? FALLING BACK TO CPU; WASB "
233                 "inference will be slow. Set indexing.wasb.device=cuda to require a
GPU.",
234                 out,
235             )
236         return True, out
237
238     def run_predict(
239         self, video_win_path: str, output_win_dir: str, video_id: Optional[str] = None
240     ) -> Optional[str]:
241         """Run WASB natively on a video. Returns the path to the trajectory CSV, or
None.
242
243         Output artifacts are namespaced by ``video_id`` (the file MD5) exactly as the
WSL
244         runner does, and the CSV is named ``{stem}_{vid12}_wasb.csv`` ? byte-for-byte
the
245         same downstream contract, so the parity comparison is apples-to-apples.
246         """
247         output_dir = os.path.abspath(output_win_dir)
248         os.makedirs(output_dir, exist_ok=True)
249         # Normalize for a cross-platform basename (tests pass Windows paths on Linux).
250         norm = str(video_win_path).replace("\\", "/")
251         stem, _ext = os.path.splitext(os.path.basename(norm))

```

```

250         # Content-derived key: same filename + different bytes -> different key (no
cache reuse).
251         key = f"{stem}__{video_id[:12]}" if video_id else (stem or "wasb")
252         out_csv = os.path.join(output_dir, f"{key}_wasb.csv")
253
254         if not self._copy_infer_script():
255             logger.error("Could not stage wasb_infer.py into the WASB repo src/.")
256             return None
257
258         weights = os.path.expanduser(self.cfg.weights_path)
259         argv = [
260             self.cfg.python_bin,
261             "wasb_infer.py",
262             "--video",
263             os.path.abspath(str(video_win_path)),
264         ]
265         frames_dir: Optional[str] = None
266         if self.cfg.stream_video:
267             # Fast path: decode in-process, no PNG extraction. Output is bit-identical
to disk.
268             argv.append("--stream-video")
269         else:
270             scratch = self.cfg.scratch_dir or output_dir
271             frames_dir = os.path.join(scratch, f"{key}_frames")
272             argv += ["--frames_out_dir", frames_dir]
273             # M4: resolve device="auto" to the effective device (cuda-if-available-else-
cpu); an
274             # explicit cuda/mps/cpu passes through unchanged. Never sends "auto" to
wasb_infer.py.
275             device = self._effective_device()
276             argv += [
277                 "--weights",
278                 weights,
279                 "--sport",
280                 self.cfg.sport,
281                 "--device",
282                 device,
283                 "--out",
284                 out_csv,
285             ]
286             # GPU-feed tuning: pass only explicit non-zero values so the default argv (and
its
287             # byte-parity guarantee) is unchanged when the knobs are unset.
288             if self.cfg.batch_size:
289                 argv += ["--batch-size", str(self.cfg.batch_size)]
290             if self.cfg.prefetch_batches:
291                 argv += ["--prefetch-batches", str(self.cfg.prefetch_batches)]
292
293             logger.info(
294                 "Running native WASB inference (device=%s%s) on %s ...",
295                 device,
296                 " [auto?fallback]" if self.cfg.device == AUTO and device != "cuda" else "",
297                 os.path.basename(norm),
298             )
299             try:
300                 res = self._run(argv, cwd=self._repo_src())
301             except FileNotFoundError:
302                 logger.error(
303                     "python binary not found: %r (set WASB_PYTHON).", self.cfg.python_bin
304                 )
305                 return None
306             except subprocess.TimeoutExpired:
307                 logger.error(
308                     "WASB native inference timed out after %ss.", self.cfg.timeout_sec
309                 )

```

```

310         return None
311     if res.returncode != 0:
312         logger.error(
313             "WASB native inference failed:\n%s",
314             (res.stderr or res.stdout or "")[-2000:],
315         )
316         return None
317     if not os.path.exists(out_csv):
318         logger.error(
319             "WASB native finished but expected CSV not found at %s", out_csv
320         )
321         return None
322     logger.info("WASB trajectory CSV: %s", out_csv)
323
324     # Success ? the CSV is the durable output; the decoded frame cache is a pure
intermediate.
325     # On any earlier failure we return above WITHOUT cleaning, so the cache survives
a resume.
326     # (Streaming never materializes frames, so there is nothing to drop on that
path.)
327     if not self.cfg.keep_frames and frames_dir:
328         logger.info(
329             "Cache hygiene: removing frame cache for %s "
330             "(set indexing.wasb.keep_frames=true to retain).",
331             key,
332         )
333         self._rm_rf(frames_dir)
334     return out_csv
335
336     # Trajectory parsing + windowing are inherited from tracknet_runner (model-
agnostic).
337     parse_trajectory_csv = staticmethod(TrackNetRunner.parse_trajectory_csv)
338     trajectory_to_action_windows = staticmethod(
339         TrackNetRunner.trajectory_to_action_windows
340     )
341

```

5. assistant (2026-07-03T19:48:29.889Z)

I have the full picture from both files. Let me look at the config models and presets to confirm the knob surface and how the runner is invoked per-job, plus check whether there's any existing warmup/persistence machinery.

6. user (2026-07-03T19:48:32.807Z)

```

1     """Pydantic models for the application configuration.
2
3     This module defines the canonical, validated, typed configuration for the
4     Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
6     Unknown / typo'd config keys are rejected at load time via extra="forbid".
7     """
8
9     from __future__ import annotations
10
11     from typing import Any, List, Literal, Optional
12
13     from pydantic import BaseModel, Field, field_validator, model_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])

```

```

21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         pts_max_gap_seconds: float = Field(
31             default=10.0,
32             ge=0.0,
33             description=(
34                 "Maximum allowed keyframe gap (seconds) before flagging as spliced. "
35                 "GoPro H.264/H.265 recordings typically have GOP intervals of 1?4s; "
36                 "a gap >10s suggests edited/missing content. Mobile phones, screen "
37                 "recordings, and other encoders may use different GOP sizes ? tune "
38                 "this threshold for your recording device."
39             ),
40         )
41         relevance: ValidationRelevanceConfig = Field(
42             default_factory=ValidationRelevanceConfig
43         )
44
45
46     class TrackNetConfig(BaseModel):
47         wsl_distro: str = "Ubuntu"
48         repo_dir: str = "~/models/TrackNetV4"
49         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
50         conda_env: str = "TrackNetV4"
51         weights_path: str = ""
52         queue_length: int = 5
53         stage_in_wsl: bool = True
54         wsl_stage_dir: str = "~/clips"
55         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
56         velocity_threshold: float = 0.02
57         # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
58         # decoded frame cache) are deleted automatically ? they are pure intermediates,
59         # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
60         # true only for debugging. On FAILURE they are always kept so a re-run can resume.
61         keep_frames: bool = False
62         # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of
63         # ~15x-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS,
64         # 0 = no extra floor (the headroom guard still applies).
65         min_free_gb: float = 0.0
66
67
68     class WasbIndexConfig(BaseModel):
69         """Typed config for the live WASB shuttle substrate (indexing.wasb).
70
71         Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
72         these knobs actually take effect ? previously this whole block was silently dropped
73         because nested config sections defaulted to extra='ignore'.
74         """
75
76         wsl_distro: str = "Ubuntu"
77         repo_dir: str = "~/models/WASB-SBDT"
78         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
79         conda_env: str = "wasb"
80         weights_path: str = (
81             "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
82         )
83         sport: str = "badminton"

```

```

84     wsl_stage_dir: str = "~/clips"
85     timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
86     velocity_threshold: float = 0.02
87     # --- Linux-native runner only (detector_impl='native'); ignored by the WSL runner.
88     # Env vars override these (WASB_DEVICE / WASB_PYTHON) for a config-free GPU box.
89     # torch device for the native runner. "auto" = cuda-if-available-else-cpu (the M4
CPU fallback
90     # for a no-GPU box); "cuda" stays strict (hard-fails without a GPU ? no silent slow-
CPU downgrade).
91     device: str = "cuda" # auto | cuda | mps | cpu
92     python_bin: str = (
93         "python" # the `wasb` conda env python (invoked by path, no `conda activate`)
94     )
95     # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
96     # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,
regenerable
97     # from the source. Set true only for debugging. On FAILURE they are kept so a re-run
resumes.
98     keep_frames: bool = False
99     # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of
the
100    # ~15x-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS,
not warns).
101    # 0 = no extra floor (the headroom guard still applies).
102    min_free_gb: float = 0.0
103    # FAST PATH ? decode the video on the fly and feed frames straight to the detector,
104    # skipping the slow per-frame PNG extraction that dominates a long clip (~46k PNGs
for a
105    # 12-min 1080p60 video, which overran the 30-min timeout). Output is bit-identical
to the
106    # PNG path (PNG is lossless) ? VERIFIED on a real GPU 2026-06-20 (native stream ==
WSL disk,
107    # 1194/1195 frames byte-identical; native run-to-run byte-identical /
deterministic).
108    # Tri-state:
109    #     None          = per-runner default ? the WSL runner keeps DISK extraction
(unchanged Windows
110    #                     behaviour); the Linux-native runner (GPU box) defaults to STREAMING
(the perf win).
111    #     True/False = force on/off for BOTH runners (e.g. set False for a container cv2
can't seek).
112    stream_video: Optional[bool] = None
113    # GPU-feed tuning for the native runner (measured on the first real L4 serving run,
114    # 2026-07-03: the default batch 8 with single-process loading left the GPU <1%
utilized ?
115    # the pipeline was CPU-feed-bound). None = wasb_infer.py's parity defaults (batch 8,
no
116    # prefetch ? byte-identical to the historical path). The cloud-serving preset opts
into
117    # the measured fast values; local/WSL behaviour is unchanged unless set explicitly.
118    batch_size: Optional[int] = None
119    # Depth of the bounded producer-thread queue that overlaps CPU-side batch assembly
120    # (decode + PIL + ToTensor) with GPU inference in streaming mode. None/0 = off.
121    prefetch_batches: Optional[int] = None
122
123
124    class FusionConfig(BaseModel):
125        """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).
126
127        Every default reproduces control behaviour: the fusion segmenter persists the
128        shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']``
is
129        true ? the person-cue features, but it does NOT gate the served handoff unless a

```

```

130     threshold is raised (`min_crossings`/`min_players` default 0 = OFF). The court
mask
131     is optional: `court_polygon=None` ? Gate B counts every detection (player-count-
only);
132     supplied ? off-court persons (spectators / adjacent courts) are excluded.
133     ""
134
135     # Which trajectory substrate seeds the windows (shuttle stays primary).
136     substrate: Literal["wasb", "tracknet"] = "wasb"
137     # Windowing velocity threshold for the fusion family (the engine reads
138     # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
139     # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
140     velocity_threshold: float = 0.02
141     # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
142     # enabled ? so the default path never imports torch / touches the GPU.
143     cue_flags: dict[str, bool] = Field(default_factory=dict)
144     # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
145     # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).
146     min_crossings: int = Field(default=0, ge=0)
147     # Served Gate B: when the person cue is on, drop windows with median in-court
players
148     # < this. 0 = OFF.
149     min_players: int = Field(default=0, ge=0)
150     # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
151     court_polygon: Optional[list[list[float]]] = None
152     # Net line for the person two-sided-motion split (person features only ? the shuttle
153     # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
154     net_axis: Literal["x", "y"] = "y"
155     net_position: float = Field(default=0.5, ge=0.0, le=1.0)
156
157
158     class IndexingConfig(BaseModel):
159         default_segmenter: str = "yolo_hybrid"
160         use_gpu: bool = True
161         # Detector backend for the trajectory hybrids (wasb/tracknet). "auto" = the real
162         # WSL runner (default, production); "stub" = a deterministic CPU mock (no GPU/WSL)
163         # so the whole pipeline is regression-testable on a CPU box / CI; "native" = the
164         # Linux-native WASB runner (no WSL/conda-activate; a GPU box ? M3.5). The
165         # RALLY_DETECTOR_IMPL env var overrides this. See
pipeline/detectors/{stub,native_wasb}_runner.py.
166         detector_impl: str = "auto"
167         motion_threshold: float = 0.08
168         min_segment_duration: float = 3.0
169         dead_time_merge_gap: float = 2.0
170         chunk_duration: float = 90.0
171         chunk_overlap: float = 10.0
172         detector_model: str = "fasterrcnn_resnet50_fpn"
173         detector_score_threshold: float = 0.5
174         yolo_velocity_threshold: float = 0.15
175         yolo_frame_skip: int = 3
176         yolo_tracker: str = "bytetrack.yaml"
177         yolo_prefetch_workers: int = 2
178         min_action_window_duration: float = 2.0
179         # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0
=a window
180         # longer than this many seconds is recursively split at its largest internal
inactivity gap (?
181         # `window_min_split_gap` s ? the most likely rally boundary), so one window can't
swallow
182         # several rallies + the dead time between them. Never merges, only cuts (recall
can't drop).
183         # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip.
Measured n=13

```

```

184      # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0,
p=0.0005; cap=6
185      # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate
0.651?0.161). ***
186      max_window_duration: float = 6.0
187      window_min_split_gap: float = (
188          0.5 # min internal gap (s) that qualifies as a split point
189      )
190      # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
191      # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
192      # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
193      # Only cuts (recall can't drop). OFF by default until measured/promoted.
194      window_hard_cap: bool = False
195      # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
196      # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
197      # |?end| gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
198      # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
199      # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant
200      # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, |?end| 4.96?3.31s (n=13).
The W1 cap
201      # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
202      window_inactivity_end: float = 1.5
203      inactivity_rally_thresh: float = 0.20
204      inactivity_min_density: float = 0.30
205      # R4 density DENOMINATOR fix (code-review finding B). The R4 trim's fast-fraction
denominator is
206      # the COUNT of active samples in the trailing window, which sparse tracking inflates
(one fast
207      # sample after a gap ? 100% dense ? tail held open). When True, the denominator is
the window's
208      # TEMPORAL length instead. No-op under dense tracking (sample-count == temporal-
span), so it only
209      # corrects the sparse-track case. *** P2b PROMOTED DEFAULT-ON (2026-06-29): golden-
set A/B at the
210      # SERVED operating point (n=15, 595 rallies) ? tolF1 0.353?0.364 (?tolF1 +0.010,
sign 10W/4L, NO
211      # regression), |?end| 2.97?2.72s, seg_count_ratio ?0.103 (p=0.006), split_rate
?0.010 (p=0.016),
212      # R1 over-seg guard PASS; served_regresses(eps=0.0)=False ?
promote_if_served_safe(structural)=True.
213      # See docs/RALLY_DETECTION_FIXES_PLAN.md §7/§8. ***
214      inactivity_temporal_density: bool = True
215      # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
216      # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
217      # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.
218      # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
219      # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
220      # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap
221      # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only

```

```

222     # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).
223     window_blob_fallback: bool = False
224     # Magnitude gate that keeps R5 surgical: only a group longer than this ×
max_window_duration is a
225     # blob (a normal long rally ~1-2× the cap is left alone; the mahadevpura-1 residual
is ~11×).
226     # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-
active
227     # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md
"R5".
228     window_blob_factor: float = 4.0
229     log_yolo_candidates: bool = True
230     store_yolo_candidates: bool = True
231     ai_handoff_padding: float = 2.0
232     ai_max_workers: int = 5
233     # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
234     # chunks of a high-bitrate (4K) source blow past the provider upload cap
235     # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
236     # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
237     # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
238     # Matches gemini_refine's --clip-scale-width default.
239     ai_chunk_scale_width: int = 1280
240     # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
241     # (wasb_hybrid / tracknet_hybrid / fusion_hybrid) skip the Phase-3 AI handoff
entirely and
242     # emit their local candidate windows AS the rallies ? no provider, no API key,
nothing
243     # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the
Gemini
244     # path, or to operate with zero cloud dependency). The pure `gemini` segmenter
ignores this.
245     skip_ai_handoff: bool = False
246     # Optional debug knob: trim every video to the first N seconds before processing.
247     debug_duration: Optional[float] = None
248
249     tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
250     wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
251     fusion: FusionConfig = Field(default_factory=FusionConfig)
252
253     @field_validator("default_segmenter")
254     @classmethod
255     def segmenter_must_be_registered(cls, v: str) -> str:
256         # Lazy import: the segmenter registry imports modules that may import
257         # config, so we resolve it at validation time rather than at module load.
258         from backend.pipeline.segmenters.base import segmenter_registry
259
260         if segmenter_registry.get(v) is None:
261             available = sorted(segmenter_registry.list_available().keys())
262             raise ValueError(
263                 f"Segmenter '{v}' is not registered. Available: {'', '.join(available)}"
264             )
265         return v
266
267     @model_validator(mode="after")
268     def _chunk_step_must_be_positive(self) -> "IndexingConfig":
269         # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
positive
270         # step makes `while t < duration: t += step` loop forever, growing memory before
any
271         # GPU work. Reject the bad config at load time instead.
272         if self.chunk_overlap >= self.chunk_duration:
273             raise ValueError(
274                 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
"
275                 f"({self.chunk_duration}); otherwise chunking never advances."

```

```

276         )
277         return self
278
279
280 class OutputConfig(BaseModel):
281     default_highlights_name: str = "highlights.mp4"
282     db_name: str = "sports_indexer.db"
283     # Directory where the DB, highlight reels, and processed clips are written.
284     # The CLI's --output-dir overrides this at startup (see backend/main.py:main).
285     output_dir: str = "output"
286
287
288 class WatermarkConfig(BaseModel):
289     """Branding overlay burned into each highlight clip during the per-clip re-encode
290     (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
291     (under `stitching.watermark` in config.json) to turn it off. The text is fully
292     configurable. The concat stage stays stream-copy (-c copy)."""
293
294     enabled: bool = True
295     text: str = "Generated by Khelsutra.guru"
296     logo_path: str = "" # optional transparent PNG overlay; "" = no logo
297     font_path: str = "" # optional .ttf; "" = use the fontconfig `font` family below
298     font: str = "Sans" # fontconfig family used when font_path is empty
299     font_size_ratio: float = Field(
300         default=0.035, gt=0.0, le=0.5
301     ) # text height as a fraction of frame height
302     opacity: float = Field(default=0.85, ge=0.0, le=1.0)
303     box: bool = True # faint backing box behind the text for legibility
304     margin: int = Field(default=24, ge=0) # px inset from the chosen corner
305     position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = (
306         "bottom-right"
307     )
308
309
310 class StitchingConfig(BaseModel):
311     begin_padding: float = 0.5
312     end_padding: float = 0.5
313     use_cache: bool = False
314     min_shot_count: int = 1
315     # Long-rally reel filter (seconds; 0 = OFF). Drop detected rally pairs whose
duration
316     # (end ? start) is below this from the highlight reel, BEFORE compiling. A cleaner
317     # reel-selection knob than `min_shot_count`: shot_count is unlabeled / "Unknown" on
the
318     # fully-local no-AI path, whereas duration is derived from the rally's own
start/end. The
319     # local detector over-fires and its false positives are mostly SHORT (motion blips,
320     # background-court fragments), so this keeps the reel to the eye-catching long
rallies.
321     # OFF by default ? the detector output is unchanged, only which rallies reach the
reel.
322     min_rally_duration: float = Field(default=0.0, ge=0.0)
323     watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
324
325
326 class LoggingConfig(BaseModel):
327     """Logging knobs for the run entrypoints (see backend/utils/logging_setup.py)."""
328
329     # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
330     # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
331     # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
332     # them to WARNING; set true to restore full chatter for request-flow debugging.
333     verbose_http: bool = False
334
335

```

```

336     class SecurityConfig(BaseModel):
337         # When set, /api/process video_path must resolve under this directory (hosted/tenant
338         # mode). Default None preserves the single-user local 'any local file' workflow.
339         allowed_video_root: Optional[str] = None
340
341         # --- Chunked upload (B2, PR-2) ---
342         # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
343         # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
344         # contain upload_dir or /api/process will reject the uploaded paths.
345         upload_dir: str = "output/uploads"
346         # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
347         # free-tier edge proxies cap request bodies there (HOSTING_PLAN $2).
348         max_upload_mb: int = 4096
349         max_part_mb: int = 95
350         allowed_upload_extensions: List[str] = ["mp4", "mov"]
351
352         # --- M9 edge auth (Cloudflare Access, D9) ---
353         # DEFAULT-OFF: edge_auth_enabled=False ? no JWT is required/verified (the local
single-user
354         # path is unchanged). When True, /api/process requires + verifies the `Cf-Access-
Jwt-Assertion`
355         # header against the team JWKS (so a direct hit to the Cloud Run URL can't spoof the
identity
356         # header) and tags the job with the verified user_id (owner_id). The team domain
(e.g.
357         # ``https://<team>.cloudflareaccess.com``) supplies the JWKS + issuer; the AUD is
the CF Access
358         # application's audience tag. NO secrets here ? these are public identifiers.
359         edge_auth_enabled: bool = False
360         cf_team_domain: str = (
361             "" # e.g. https://<team>.cloudflareaccess.com (issuer + /certs JWKS)
362         )
363         cf_access_aud: str = "" # the CF Access application AUD tag the token must carry
364         # (CORS allow-origins is set via the RALLY_CORS_ALLOW_ORIGINS env var ? read at
import in
365         # backend/main.py so importing the app does no filesystem I/O; default [""].
Credentials stay
366         # OFF ? Cf-Access is a header, not a cookie.)
367
368
369     class StorageConfig(BaseModel):
370         """Pluggable storage backend (docs/DECOUPLED_COMPUTE/). Default ``local`` = today's
371         behaviour (filesystem paths, byte-for-byte). ``s3`` covers AWS + Cloudflare R2 + D0
Spaces
372         + MinIO (via ``endpoint_url``); ``gcs`` = Google Cloud Storage; ``gdrive`` = a
locally MOUNTED
373         Google Drive (``gdrive.mount_root``). Per-backend settings are loose dicts passed to
the constructor ?
374         **secrets are never stored here**, only endpoints / regions / file paths / source-
selectors
375         (boto3/google auth chains supply credentials). See ``backend/storage/``. """
376
377         backend: Literal["local", "s3", "gcs", "gdrive", "gdrive-api"] = "local"
378         # Output root for produced artifacts. "" => fall back to output.output_dir (non-
breaking).
379         output_root: str = ""
380         # --- Input side (COMPUTE_DECOUPLED_SERVING M2; parallel to the output
`backend`/`output_root`).
381         # Default ``local`` / "" keeps today's behaviour: a bare path / ``local://`` URI is
read IN PLACE
382         # (identity, no copy). Set these to read inputs from a bucket / mounted Drive ? a
bare/relative
383         # input name is resolved against ``input_root`` (e.g.
``input_root``="gs://bucket/videos"`` + "x.mp4"
384         # => ``gs://bucket/videos/x.mp4``) or, if ``input_root`` is empty, prefixed with

```

```

`input_backend`
385     # (`"gcs"` + "bucket/x.mp4" => `gcs://bucket/x.mp4`). An explicit scheme on the
input
386     # (`gs://?`/`s3://?`) or an absolute local path ALWAYS wins. Non-local inputs
are fetched to a
387     # scratch dir before processing (see `backend.storage.acquire_local_input`).
388     input_backend: Literal["local", "s3", "gcs", "gdrive", "gdrive-api"] = "local"
389     input_root: str = ""
390     # --- Per-video workspace (docs/PER_VIDEO_WORKSPACE.md) ---
391     # One folder per video keyed by the MD5 video_id:
<root>/[<workspace_prefix>]/<video_id>/...
392     # so a local deploy and a cloud (bucket) deploy share ONE relative layout (only the
root
393     # differs). DEFAULT-OFF: convergence is phased + flipped via a Launch Report (D5).
When OFF
394     # the legacy flat layout is used unchanged.
395     single_folder_per_video: bool = False
396     # Durable workspace root URI. "" => precedence: output_root, then output.output_dir.
397     workspace_root: str = ""
398     # Cloud namespace prefix under the root so per-video folders sit tidily beside
399     # videos/ labels/ weights/ (e.g. gs://bucket/workspaces/<video_id>/). "" = root
directly.
400     workspace_prefix: str = "workspaces"
401     local: dict[str, Any] = Field(default_factory=dict)
402     s3: dict[str, Any] = Field(default_factory=dict)
403     gcs: dict[str, Any] = Field(default_factory=dict)
404     gdrive: dict[str, Any] = Field(default_factory=dict)
405     # gdrive-api settings (e.g. {"root_folder_id": "<id>"}); read via the
hyphen?underscore lookup in
406     # backend.storage.get_storage. NO secrets ? creds come from ADC over the Drive
scope.
407     gdrive_api: dict[str, Any] = Field(default_factory=dict)
408
409
410     class DeploymentConfig(BaseModel):
411         """Deployment profile ? one switch that selects a coherent bundle of compute +
storage
412         knobs (COMPUTE_DECOUPLED_SERVING M1, `docs/COMPUTE_DECOUPLED_SERVING/README.md`
§4.3).
413
414         **Pure / additive:** the empty default profile (`""`) applies NO preset, so the
pipeline
415         behaves exactly as before this section existed. A profile is opt-in (`config.json`
`deployment.profile` or the `DEPLOYMENT_PROFILE` env var); auto-detect only ever
416         *suggests* one (D15 ? cloud spend is never entered silently). Nothing in the core
reads
417         these fields yet (M1): selecting a profile works purely by pre-setting the EXISTING
knobs
418         (`indexing.detector_impl` / `skip_ai_handoff`, `storage.backend`) in the
loader.
419
420         Valid values mirror `backend/config/presets.py` (parity-pinned by a unit test)."""
421
422         # "" = no profile (today's manual knobs). Others apply presets.PROFILE_PRESETS.
423         profile: Literal["", "truly-local", "cloud-serving", "cpu-mock"] = ""
424         # Where the core runs. Set by the profile preset; overridable in config.json. Inert
in M1
425
426         # (recorded for observability; consumed by the ComputeBackend seam from M2+).
427         compute_target: Literal["", "local_gpu", "cloud_run", "cpu_mock"] = ""
428
429
430     class BillingConfig(BaseModel):
431         """Per-job cost ledger (COMPUTE_DECOUPLED_SERVING M8, D8).
432
433         **DEFAULT-OFF:** `enabled=False` ? nothing records cost (the v6 cost columns stay

```

```

NULL =
434         today's behaviour). When enabled, a job's cost is computed + stored in **USD** via
the versioned
435         ``pricebook`` DB table and **displayed in INR** at ``fx_inr_per_usd``. The default
pricebook
436         version is the seeded ``placeholder-v0`` (every rate ``0.0`` ? cost ``0``) until the
owner
437         inserts a real, calibrated version and points ``pricebook_version`` at it.
438
439         ? Even with ``enabled=True`` + a real pricebook, recorded cost stays **0** until a
follow-up
440         wires the worker to emit per-job ``usage`` (gpu/wall/egress/gemini) ? M8 lands the
ledger +
441         pricebook + the recording seam; usage emission (the metering hooks) is a separate
step.
442
443         enabled: bool = False
444         pricebook_version: str = (
445             "placeholder-v0" # the seeded $0 version; owner flips to a real one
446         )
447         fx_inr_per_usd: float = 83.0 # USD?INR display rate (Bangalore market)
448         margin: float = 0.0 # resale markup (?0); 0 = bill at cost
449
450
451     class FlywheelConfig(BaseModel):
452         """Data-flywheel harness (COMPUTE_DECOUPLED_SERVING M11, D12) ? the
consent?capture?eval?goldenize
453         loop: on a CONSENTED adult job, durably **capture** the source + rally output into
the per-video
454         workspace ? **shadow-eval** owned-vs-Gemini ? **propose promotion** to the golden
TRAINING set, so
455         the owned model climbs toward the D11 ship floor.
456
457         **DEFAULT-OFF + INERT:** ``enabled=False`` ? nothing is captured/evaluated (byte-
identical to
458         today). ? **Consent is FAKED-DENY until counsel + the consent schema land:** even
``enabled=True``
459         captures **nothing**, because ``flywheel.consent_attested`` currently returns
``False`` for every
460         job (the privacy guardrail ? D10: adults-only, explicit opt-in, derived-data + auto-
delete-row).
461         This is the PLUMBING only; the real consent gate (counsel-reviewed ToS/DPA ? consent
columns ?
462         wiring) + the owned-model + golden-label data needed to *activate* shadow-
eval/promotion are
463         owner/counsel/GPU scope. Pairs with M9-consent.
464
465         enabled: bool = False
466         workspace_prefix: str = (
467             "flywheel" # per-video workspace prefix for captured artifacts
468         )
469         shadow_eval: bool = True # run the owned-vs-Gemini shadow comparison on capture
470         min_promote_n: int = 5 # promote_if_served_safe `min_n` (the served-safe gate)
471
472
473     class AppConfig(BaseModel):
474         """Root configuration object.
475
476         All runtime configuration should be accessed via an instance of this class
477         (or the loader) rather than raw dict lookups.
478
479
480         sport: Sport = "badminton"
481         ai_provider: str = "gemini"
482         # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable

```

```

flagship;
483      # the alias never deprecates underneath us. Eval reports record the run date
alongside.
484      ai_model: str = "gemini-flash-latest"
485
486      validation: ValidationConfig = Field(default_factory=ValidationConfig)
487      indexing: IndexingConfig = Field(default_factory=IndexingConfig)
488      output: OutputConfig = Field(default_factory=OutputConfig)
489      stitching: StitchingConfig = Field(default_factory=StitchingConfig)
490      security: SecurityConfig = Field(default_factory=SecurityConfig)
491      logging: LoggingConfig = Field(default_factory=LoggingConfig)
492      storage: StorageConfig = Field(default_factory=StorageConfig)
493      deployment: DeploymentConfig = Field(default_factory=DeploymentConfig)
494      billing: BillingConfig = Field(default_factory=BillingConfig)
495      flywheel: FlywheelConfig = Field(default_factory=FlywheelConfig)
496
497      # Reject unknown keys ? typed config migration is complete (D3).
498      model_config = {
499          "extra": "forbid",
500          "validate_assignment": True,
501      }
502
503
504      # Convenience type alias for callers that want to annotate
505      Config = AppConfig
506

```

7. user (2026-07-03T19:48:33.971Z)

```

1      """Deployment-profile presets + environment auto-detection (COMPUTE_DECOUPLED_SERVING
M1).
2
3      A *deployment profile* is a single switch that applies a coherent BUNDLE of existing
4      config knobs, so the truly-local / cloud-serving / cpu-mock modes don't drift apart one
5      knob at a time (the design in ``docs/COMPUTE_DECOUPLED_SERVING/README.md`` §4.3).
6
7      Precedence (lowest ? highest), realised by the loader::
8
9          built-in model defaults < profile preset < config.json < env < worker --set
10
11      This module is **pure data + pure functions** ? no IO, and (critically) **no torch
import at
12      module load**. The loader applies a preset ONLY when a profile is EXPLICITLY selected
13      (``config.json`` ``deployment.profile`` or the ``DEPLOYMENT_PROFILE`` env var); with no
14      profile selected the loader is byte-identical to the pre-M1 behaviour (``profile=""`` ==
15      today). Auto-detect only ever **suggests** a profile ? it never auto-applies one (D15:
cloud
16      spend is never entered silently).
17      """
18
19      from __future__ import annotations
20
21      import copy
22      import os
23      from typing import Any, Mapping, Optional
24
25      # Canonical enum values, defined ONCE here. ``models.DeploymentConfig``'s ``Literal``
mirrors
26      # these and ``tests/test_config_deployment.py`` pins parity so the two can never drift.
27      PROFILES: tuple[str, ...] = ("truly-local", "cloud-serving", "cpu-mock")
28      COMPUTE_TARGETS: tuple[str, ...] = ("local_gpu", "cloud_run", "cpu_mock")
29
30      # Each profile maps 1:1 to its canonical compute_target. Used to apply the preset and to
warn
31      # when config.json explicitly overrides compute_target into a state inconsistent with

```

the profile.

```
32     COMPUTE_TARGET_FOR_PROFILE: dict[str, str] = {
33         "truly-local": "local_gpu",
34         "cloud-serving": "cloud_run",
35         "cpu-mock": "cpu_mock",
36     }
37
38     # A preset is a partial, nested override dict over the model defaults. It touches ONLY
knobs
39     # that EXIST today ? M1 is pure/additive: the new
``deployment.{profile,compute_target}`` plus
40     # ``indexing.{detector_impl,skip_ai_handoff}`` and ``storage.backend``. Later milestones
extend
41     # these bundles ? ``input_backend`` (M2) and the configurable output/workspace base (M3,
42     # Launch-Report-gated) ? deliberately NOT set here so M1 never pre-empts a gated flip.
43     PROFILE_PRESETS: dict[str, dict[str, Any]] = {
44         "truly-local": {
45             "deployment": {"profile": "truly-local", "compute_target": "local_gpu"},
46             # detector_impl stays "auto" (the runner picks WSL vs native by env on the GPU
box).
47             # Zero-cloud / privacy-first ? run fully local with no Gemini handoff.
48             "indexing": {"skip_ai_handoff": True},
49             "storage": {"backend": "local"},
50         },
51         "cloud-serving": {
52             "deployment": {"profile": "cloud-serving", "compute_target": "cloud_run"},
53             # Linux L4 (cuda); Gemini handoff ON until the owned model clears the #172 floor
(D11).
54             # wasb GPU-feed tuning: the parity defaults (batch 8, no prefetch) left the paid
L4
55             # <1% utilized on the first real serving run (2026-07-03) ? the pipeline was
56             # CPU-feed-bound. Serving opts into the measured fast values; local/WSL
unchanged.
57             "indexing": {
58                 "detector_impl": "native",
59                 "skip_ai_handoff": False,
60                 "wasb": {"batch_size": 64, "prefetch_batches": 4},
61             },
62             "storage": {"backend": "gcs"},
63         },
64         "cpu-mock": {
65             "deployment": {"profile": "cpu-mock", "compute_target": "cpu_mock"},
66             # Deterministic CPU mock ? no GPU/WSL, no provider/API. The CI / regression
profile.
67             "indexing": {"detector_impl": "stub", "skip_ai_handoff": True},
68             "storage": {"backend": "local"},
69         },
70     }
71
72
73     def is_known_profile(profile: str) -> bool:
74         """True for a non-empty, recognised profile name."""
75         return bool(profile) and profile in PROFILE_PRESETS
76
77
78     def preset_for(profile: str) -> dict[str, Any]:
79         """Return a deep COPY of the preset bundle for ``profile`` (so callers can't mutate
the
80         module-level table), or ``{}`` for ``""`` / an unknown profile."""
81         return copy.deepcopy(PROFILE_PRESETS.get(profile, {}))
82
83
84     def _truthy(val: Optional[str]) -> bool:
85         """Common env-var truthiness (``1/true/yes/on``, case/space-insensitive)."""
86         if val is None:
```

```

87         return False
88     return val.strip().lower() in {"1", "true", "yes", "on"}
89
90
91     def _cuda_available() -> bool:
92         """Best-effort CUDA probe. torch is heavy + optional, so it is imported lazily here
93         and
94         ONLY when a caller opts into ``probe_cuda``; any import/runtime failure means 'no
95         GPU'."""
96         try:
97             # Intentionally lazy + local: keeps the config-load path torch-free (torch is
98             heavy
99             # and optional). Only reached when a caller opts into probe_cuda.
100             import torch
101
102             return bool(torch.cuda.is_available())
103         except Exception:
104             return False
105
106
107     def probe_cuda_available() -> bool:
108         """Public best-effort CUDA probe (lazy torch import; ``False`` on any failure).
109         Exposed for the per-profile startup checks (``backend/config/startup.py``) so a
110         caller
111         that *wants* the GPU heads-up can pass the result in without the startup module
112         importing
113         torch itself. Best-effort by design: if torch lives only in a detector subprocess
114         env
115         (native WASB), this returns ``False`` in the main process ? which is why the startup
116         checks
117         treat a GPU mismatch as a WARNING (the runner healthcheck is the authority), never a
118         hard
119         error that could false-fail."""
120         return _cuda_available()
121
122
123     def suggest_profile(
124         env: Optional[Mapping[str, str]] = None, *, probe_cuda: bool = False
125     ) -> str:
126         """SUGGEST (never auto-apply ? D15) a deployment profile from the environment.
127
128         Signals, in priority order:
129
130         - ``K_SERVICE`` set (Cloud Run)           ? ``"cloud-serving"``
131         - ``CI`` truthy                             ? ``"cpu-mock"``
132         - ``probe_cuda`` and a CUDA GPU present ? ``"truly-local"``
133         - otherwise                                 ? ``""`` (no strong suggestion)
134
135         ``probe_cuda`` defaults to ``False`` so the common config-load path stays cheap and
136         never
137         imports torch; a CLI ``--detect`` or cloud-serving startup that wants the GPU probe
138         passes
139         ``probe_cuda=True``. The result is advisory only: the caller decides whether to act
140         on it.
141
142         """
143         e = os.environ if env is None else env
144         if e.get("K_SERVICE"):
145             return "cloud-serving"
146         if _truthy(e.get("CI")):
147             return "cpu-mock"
148         if probe_cuda and _cuda_available():
149             return "truly-local"
150         return ""

```

8. user (2026-07-03T19:48:34.679Z)

```
1      # Cloud Run GPU **Job** image for the rally worker (COMPUTE_DECOUPLED_SERVING M6/M7,
ADR-012/D16).
2      #
3      # Runs `python -m backend.worker infer ?` INLINE on an L4 (detect + window + stitch co-
located so
4      # CPU/wall + egress are metered into one job ? D4). Built + pushed + verified by the
owner on the
5      # M7 real run (the $100/?10k budget, D17); CI does NOT build this (a CUDA image is too
heavy) ? the
6      # dispatch shim is proven with a fake client (tests/test_compute_backend.py).
7      #
8      # Two python stacks, deliberately separate (a framework conflict otherwise ? same split
as the
9      # native GPU box, deploy/digitalocean/gpu_setup.sh):
10     # * the MAIN app env (this repo + ffmpeg) drives the pipeline + stitching;
11     # * a conda `wasb` env (py3.8 + torch 1.11.0+cu113) runs the WASB shuttle detector as
a subprocess
12     # (the native runner shells out to $WASB_PYTHON), so the app's torch never co-
installs with it.
13     #
14     # ? WASB-C3: the pretrained badminton weights are academic-data-trained ? provenance NOT
cleared for
15     # commercial sharing. They are NOT baked into this image; stage them at runtime (a
bucket fetch or a
16     # mount) and point $WASB_WEIGHTS at them. Resolve WASB-C3 before any public, non-owner
serving.
17
18     FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04
19
20     ENV DEBIAN_FRONTEND=noninteractive \
21         PYTHONUNBUFFERED=1 \
22         PIP_NO_CACHE_DIR=1 \
23         CONDA_DIR=/opt/conda \
24         WASB_REPO_DIR=/opt/models/WASB-SBDT
25
26     # ffmpeg (stitch/decode) + DejaVu fonts (the watermark filter) + git/curl for the WASB
env build.
27     RUN apt-get update && apt-get install -y --no-install-recommends \
28         ffmpeg fonts-dejavu-core git curl ca-certificates build-essential \
29         python3 python3-pip python3-venv \
30         && rm -rf /var/lib/apt/lists/*
31
32     # --- WASB detector env (micromamba + conda-forge ? BSD-licensed, NO Anaconda ToS; P3a,
ADR-005) ---
33     # Enforces ADR-005 (every dependency ? code, data, weights, AND the build toolchain ?
stays
34     # MIT/Apache/BSD): micromamba (BSD-3) + conda-forge (community) replaces Miniconda's
Anaconda-ToS-gated
35     # `defaults` channels (the ?200-employee Terms-of-Service trap flagged in
PACKAGING_PLAN.md §3).
36     # MAMBA_ROOT_PREFIX=$CONDA_DIR (= /opt/conda) keeps the env at $CONDA_DIR/envs/wasb so
$WASB_PYTHON below
37     # is UNCHANGED. Same verified torch-1.11.0+cu113 WASB stack; the pip wheels are
unaffected by the switch.
38     ENV MAMBA_ROOT_PREFIX=/opt/conda
39     RUN curl -sSL https://micro.mamba.pm/api/micromamba/linux-64/latest \
40         | tar -xj -C /usr/local bin/micromamba \
41         && micromamba create -y -q -n wasb -c conda-forge python=3.8 \
42         && git clone --depth 1 https://github.com/nttcom/WASB-SBDT.git "$WASB_REPO_DIR" \
43         && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
44             torch==1.11.0+cu113 torchvision==0.12.0+cu113 --extra-index-url
https://download.pytorch.org/whl/cu113 \
45         && ( [ -f "$WASB_REPO_DIR/requirements.txt" ] && "$CONDA_DIR/envs/wasb/bin/python"
```

```

-m pip install -q -r "$WASB_REPO_DIR/requirements.txt" || true ) \
46      && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
47      hydra-core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml
matplotlib "numpy<1.24"
48
49      # --- Main application env ---
50      WORKDIR /app
51      COPY pyproject.toml README.md ./
52      COPY requirements-trackers.txt ./
53      COPY backend ./backend
54      COPY training ./training
55      # Install with the [storage-gcs] extra: the cloud worker PULLS the gs:// input and
PUSHES the
56      # gs:// outputs/done-marker through backend.storage (google-cloud-storage is lazy-
imported there).
57      # Without it the job dies at the first gs:// fetch with `ModuleNotFoundError:
google.cloud` ? the
58      # real-Cloud-Run failure the mock-client tests can't see (the bucket I/O is faked in
tests).
59      # D13/P12: `trackers` (ByteTrack's new home) is a git-only dep (broken PyPI wheel ?
60      # roboflow/trackers#474) so it's NOT pulled by `pip install .`; install it out-of-band
before
61      # the main env, or the person-cue path crashes with ModuleNotFoundError. The immutable
git SHA is
62      # single-sourced in requirements-trackers.txt.
63      RUN python3 -m pip install --upgrade pip \
64      && python3 -m pip install -r requirements-trackers.txt \
65      && python3 -m pip install ".[storage-gcs]"
66
67      # The native runner finds the WASB env + repo by env (no `conda activate` needed);
weights come in at
68      # runtime (WASB-C3). detector_impl=native = the Linux-native WASB runner; cuda stays the
default.
69      # PYTHONUNBUFFERED: without it the worker's stdout/stderr sit in the block buffer until
process
70      # exit, so Cloud Logging shows a single end-of-run burst instead of live progress
(observed on the
71      # first real M7 runs, 2026-07-03) ? a running job looks stalled and mid-run failures log
nothing.
72      ENV PYTHONUNBUFFERED=1 \
73      WASB_PYTHON=/opt/conda/envs/wasb/bin/python \
74      WASB_REPO_DIR=/opt/models/WASB-SBDT \
75      WASB_WEIGHTS=/opt/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar \
76      RALLY_DETECTOR_IMPL=native
77
78      # The dispatcher (CloudRunCompute) supplies the per-execution args:
79      # infer --video-uri gs:/// --out-uri gs:/// --done-uri gs:/// --segmenter wasb_hybrid
--set ?
80      ENTRYPOINT ["python3", "-m", "backend.worker"]
81

```

9. assistant (2026-07-03T19:48:39.508Z)

I now have the complete picture. Let me confirm two specifics: (1) whether `model.eval()` / `grad state` is set inside `build_detector` / `run_tensor` (which I cannot read, but can reason about from the WASB call surface and the `torch.no_grad()` scoping in `wasb_infer.py`), and (2) whether the worker/dispatch uses a Cloud Run Job (ephemeral, one job per video) vs a Service. Let me grep the worker and compute backend for the job lifecycle.

10. user (2026-07-03T19:48:43.856Z)

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\config\models.py:243: # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the Gemini

```
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\ui\assets\index-Cqr7zJeJ.js:60:[Omitted long matching line]
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\person_detector.py:102:         self.model.eval()
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\person_detector.py:152:         with torch.no_grad():
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\wasb_infer.py:619:
stack.enter_context(torch.no_grad())
```

11. user (2026-07-03T19:48:45.612Z)

Found 32 files

```
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\config\presets.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\native_wasb_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\config\models.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\worker\__main__.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\config\startup.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\utils\run_telemetry.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\utils\app_home.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\ui\assets\index-Cqr7zJeJ.js
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\reporting\spec.yaml
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\results.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\providers\gemini.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\segmenters\base.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\segmenters\tracknet_hybrid.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\segmenters\trajectory_hybrid.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\segmenters\wasb_hybrid.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\stub_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\tracknet_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\wasb_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\main.py
```

```

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\orchestration.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\base.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\device.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\infrastructure\execution_policy.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\infrastructure\database_jobs.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\infrastructure\database.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\config\loader.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\compute\cloud_run.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\compute\__init__.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\compute\base.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\api\static\app.js
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\api\job_worker.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\annotations\base.py

```

12. assistant (2026-07-03T19:48:49.467Z)

The grep confirms the WASB detector's `build\_detector`/`run\_tensor` are opaque (in the WASB repo, not readable), and the only grad/eval control in wasb_infer.py is the `torch.no\_grad()` at line 619. Let me confirm the Cloud Run Job (ephemeral, one-execution-per-video) lifecycle in the compute backend and worker entry.

13. user (2026-07-03T19:48:51.038Z)

```

1      """`CloudRunCompute` ? dispatch one inference job to a Cloud Run GPU **Job** and
bucket-poll
2      for its completion (COMPUTE_DECOUPLED_SERVING M6, decisions D16 + D7).
3
4      Flow (the cloud-serving compute path):
5      1. **Dispatch** a Cloud Run Job execution whose container runs `python -m
backend.worker infer
6      --video-uri <gs> --out-uri <gs> --done-uri <gs> ?` (the SAME worker, on the L4
GPU).
7      2. **Bucket-poll** `done_uri` via `execution_policy.poll_until` + the storage
layer (D16 ? the
8      completion signal is a tiny marker object the worker writes, NOT the Cloud Run API
status, so
9      it is storage-backed and reuses the existing park/resume primitives). Scale-to-zero
(D7): the
10     cold-start is absorbed by the async poll. On a poll TIMEOUT (`max_wait_sec` can
undercut the
11     job's own `--task-timeout` budget) the shim does ONE final marker check ? a slow-
but-healthy
12     worker that finished just past the deadline is rescued, not wasted ? then best-
effort CANCELS
13     the still-running execution so the L4 doesn't keep billing until `--task-
timeout`.

```

```

14     3. **Read** the marker (it carries ``video_id`` + the worker's returncode) ? fetch
15     ``outputs/<video_id>/report.json`` ? ``ComputeResult``.
16
17     The dispatched container is forced to run **INLINE** (``compute_target=local_gpu`` +
18     native detector)
19     so it never recursively re-dispatches ? the dispatcher decides cloud, the container just
20     computes.
21
22     The Cloud Run trigger is an **injected client** (``CloudRunJobClient``) so the whole
23     shim is unit-tested
24     with a fake (no GCP). The real ``GoogleCloudRunJobClient`` lazy-imports ``google-cloud-
25     run`` and is
26     owner-verified on the M7 real run.
27     ""
28
29     from __future__ import annotations
30
31     import json
32     import logging
33     import os
34     import tempfile
35     import uuid
36     from abc import ABC, abstractmethod
37     from typing import Any, List, Optional
38
39     from backend.compute.base import ComputeBackend, ComputeJobSpec, ComputeResult
40     from backend.infrastructure.execution_policy import (
41         DEFAULT_POLICY,
42         ExecutionPolicy,
43         PolicyTimeout,
44         poll_until,
45     )
46     from backend.storage import fetch, get_storage, parse_uri
47
48     LOG = logging.getLogger("compute.cloud_run")
49
50     # The container worker must run INLINE on the GPU ? never re-enter the cloud-dispatch
51     branch.
52     # (compute_target=local_gpu disarms the cmd_infer dispatch guard; native = the L4 WASB
53     runner.)
54     _DEFAULT_CONTAINER_OVERRIDES: tuple[str, ...] = (
55         "deployment.compute_target=local_gpu",
56         "indexing.detector_impl=native",
57     )
58
59     class CloudRunJobClient(ABC):
60         """Triggers a Cloud Run **Job** execution with per-execution container arg
61         overrides.
62
63         Abstracted so ``CloudRunCompute`` is testable with a fake (no GCP). A real
64         implementation
65         overrides the job's container ``args`` for this execution and starts it."""
66
67         @abstractmethod
68         def run_job(
69             self,
70             job_name: str,
71             argv: List[str],
72             *,
73             region: str,
74             project: Optional[str] = None,
75         ) -> str:
76             """Start a Cloud Run Job execution running ``python -m backend.worker <argv>``;
77             return an

```

```

70         execution id/name (for logs). Does NOT block on completion (the caller bucket-
polls)."""
71
72     @abstractmethod
73     def cancel_execution(self, execution: str) -> None:
74         """Cancel a running execution by the name ``run_job`` returned. Called when the
bucket-poll
75         times out so the abandoned execution doesn't keep billing the GPU until
``--task-timeout``.
76         May raise ? the caller treats every failure as best-effort (the original poll
timeout is
77         NEVER masked by a cancel error)."""
78
79
80     class GoogleCloudRunJobClient(CloudRunJobClient):
81         """Real client ? lazy-imports ``google-cloud-run``. Owner-verified on the M7 real
run; CI uses
82         a fake, so this code path is never exercised in tests (the SDK isn't a test
dependency)."""
83
84         def run_job(
85             self,
86             job_name: str,
87             argv: List[str],
88             *,
89             region: str,
90             project: Optional[str] = None,
91         ) -> str:
92             # A None/empty project would build a bogus "projects/None/locations/..." path
(job_path
93             # requires a real str). Fail loudly BEFORE the SDK import so the misconfig is
exercisable
94             # in CI (where google-cloud-run is absent) and so mypy narrows project to str
for job_path.
95             if not project:
96                 raise RuntimeError(
97                     "GOOGLE_CLOUD_PROJECT is unset ? the real Cloud Run dispatch needs a GCP
project id "
98                     "to resolve the job path; set it on the control plane (see
get_compute_backend)."
99                 )
100             try:
101                 from google.cloud import run_v2 # type: ignore[attr-defined]
102             except Exception as exc: # pragma: no cover - real SDK not present in CI
103                 raise RuntimeError(
104                     "google-cloud-run is required for CloudRunCompute's real client; install
it on the "
105                     "control plane (it is intentionally NOT a CI/test dependency)."
106                 ) from exc
107             client = (
108                 run_v2.JobsClient()
109             ) # pragma: no cover - exercised only on the real M7 run
110             name = client.job_path(project, region, job_name)
111             overrides = run_v2.RunJobRequest.Overrides(
112                 container_overrides=[
113                     run_v2.RunJobRequest.Overrides.ContainerOverride(args=argv)
114                 ]
115             )
116             op = client.run_job(
117                 request=run_v2.RunJobRequest(name=name, overrides=overrides)
118             )
119             return (
120                 getattr(op, "metadata", None)
121                 and getattr(op.metadata, "name", "")
122                 or "execution"

```

```

123         )
124
125     def cancel_execution(self, execution: str) -> None:
126         # run_job falls back to the literal string "execution" when the operation
127         # no name ? that is NOT a resolvable resource name. Fail loudly BEFORE the SDK
128         # (mirrors run_job's project guard, #342) so the misuse is exercisable in CI
129         # google-cloud-run is absent).
130         if "/executions/" not in (execution or ""):
131             raise RuntimeError(
132                 f"not a fully-qualified Cloud Run execution name: {execution!r} ?
133                 "projects/<p>/locations/<l>/jobs/<j>/executions/<e> (run_job could not
134                 "one from the operation metadata, so there is nothing to cancel by
135                 )
136         try:
137             from google.cloud import run_v2 # type: ignore[attr-defined]
138         except Exception as exc: # pragma: no cover - real SDK not present in CI
139             raise RuntimeError(
140                 "google-cloud-run is required to cancel a Cloud Run execution; install
141                 "control plane (it is intentionally NOT a CI/test dependency)."
142             ) from exc
143         client = (
144             run_v2.ExecutionsClient()
145         ) # pragma: no cover - exercised only on a real run
146         client.cancel_execution(request=run_v2.CancelExecutionRequest(name=execution))
147
148
149     class CloudRunCompute(ComputeBackend):
150         """Dispatch to a Cloud Run GPU Job + bucket-poll for completion."""
151
152     def __init__(
153         self,
154         *,
155         run_client: CloudRunJobClient,
156         job_name: str,
157         region: str,
158         project: Optional[str] = None,
159         storage_config: Optional[dict] = None,
160         policy: ExecutionPolicy = DEFAULT_POLICY,
161         token: Optional[str] = None,
162         container_overrides: Optional[tuple[str, ...]] = None,
163     ) -> None:
164         self._client = run_client
165         self._job_name = job_name
166         self._region = region
167         self._project = project
168         self._storage_config = storage_config or {}
169         self._policy = policy
170         # The dispatch token keys the done-marker so concurrent jobs to one bucket don't
171         # None ? a fresh uuid4 per run_infer call (prod); injected ? deterministic
172         self._token = token
173         self._container_overrides = (
174             _DEFAULT_CONTAINER_OVERRIDES
175             if container_overrides is None
176             else tuple(container_overrides)
177         )
178

```

```

179 def run_infer(self, spec: ComputeJobSpec) -> ComputeResult:
180     out_root = spec.out_uri.rstrip("/")
181     token = self._token or uuid.uuid4().hex
182     done_uri = f"{out_root}/_dispatch/{token}.done"
183     argv: List[str] = [
184         "infer",
185         "--video-uri",
186         spec.video_uri,
187         "--out-uri",
188         spec.out_uri,
189         "--done-uri",
190         done_uri,
191     ]
192     if spec.segmenter:
193         argv += ["--segmenter", spec.segmenter]
194     for kv in (*spec.config_overrides, *self._container_overrides):
195         argv += ["--set", kv]
196
197     execution = self._client.run_job(
198         self._job_name, argv, region=self._region, project=self._project
199     )
200     LOG.info(
201         "cloud-run: dispatched job=%s exec=%s; bucket-polling %s",
202         self._job_name,
203         execution,
204         done_uri,
205     )
206
207     try:
208         marker = poll_until(
209             lambda: self._read_marker(done_uri),
210             self._policy,
211             label="cloud-run-infer",
212             logger=LOG,
213         )
214     except PolicyTimeout as timeout_exc:
215         marker = self._handle_poll_timeout(done_uri, str(execution), timeout_exc)
216         video_id = str(marker.get("video_id", ""))
217         # A MISSING returncode defaults to FAILURE (1), never success ? a present-but-
unreadable /
218         # garbled marker must not masquerade as rc=0. (The worker always writes both
fields.)
219         rc = int(marker.get("returncode", 1))
220         if not video_id:
221             LOG.warning(
222                 "cloud-run: job=%s completion marker present but empty/unreadable (no "
223                 "video_id) ? treating as failure",
224                 self._job_name,
225             )
226             rc = rc or 1
227         outputs_uri = f"{out_root}/outputs/{video_id}/" if video_id else ""
228         report = self._read_json(f"{outputs_uri}report.json") if outputs_uri else {}
229         LOG.info(
230             "cloud-run: job=%s done (video_id=%s rc=%d)", self._job_name, video_id, rc
231         )
232         return ComputeResult(
233             returncode=rc,
234             outputs_uri=outputs_uri,
235             video_id=video_id,
236             report=report,
237             execution=str(execution or ""),
238         )
239
240 def _handle_poll_timeout(
241     self, done_uri: str, execution: str, timeout_exc: PolicyTimeout

```

```

242         ) -> dict:
243             """Poll deadline hit ? rescue a just-finished result, else cancel + re-raise.
244
245             The poll budget (`max_wait_sec`, default 1800s) can undercut the deployed
job's own
246             ``--task-timeout`` (3600s), so a slow-but-healthy worker may finish just past
the deadline:
247             ONE final marker check rescues that completed GPU run instead of wasting it. If
the marker
248             is genuinely absent, the execution is still running ? best-effort cancel it
(otherwise the
249             L4 keeps billing until ``--task-timeout``) and re-raise a ``PolicyTimeout``
carrying the
250             execution name so the operator can verify the state in the GCP console. A cancel
failure
251             NEVER masks the original timeout."""
252             try:
253                 late = self._read_marker(done_uri)
254             except Exception: # noqa: BLE001 ? the final check is best-effort too: absence
? failure
255                 LOG.warning(
256                     "cloud-run: final post-timeout marker check failed ? treating as
absent",
257                     exc_info=True,
258                 )
259                 late = None
260             if late is not None:
261                 LOG.warning(
262                     "cloud-run: job=%s exec=%s done-marker found on the final post-timeout
check "
263                     "? rescuing the completed result",
264                     self._job_name,
265                     execution,
266                 )
267                 return late
268             try:
269                 self._client.cancel_execution(execution)
270             except Exception as cancel_exc: # noqa: BLE001 ? best-effort: never mask the
timeout
271                 cancel_note = f"best-effort cancel FAILED ({cancel_exc})"
272                 LOG.warning(
273                     "cloud-run: best-effort cancel of execution=%s failed: %s",
274                     execution,
275                     cancel_exc,
276                     exc_info=True,
277                 )
278             else:
279                 cancel_note = "cancel requested"
280                 LOG.info("cloud-run: requested cancel of execution=%s", execution)
281             raise PolicyTimeout(
282                 f"{timeout_exc} [cloud-run job={self._job_name} "
283                 f"execution={execution or '<unknown>'}; {cancel_note} ? verify the execution
"
284                 "state in the GCP console]"
285             ) from timeout_exc
286
287         # --- storage-backed completion signal (bucket-poll) ---
288         def _read_marker(self, done_uri: str) -> Optional[dict]:
289             """The done-marker dict if the worker has written it, else None ? poll_until
keeps waiting."""
290             ref = parse_uri(done_uri)
291             if not get_storage(ref.backend, config=self._storage_config).exists(ref):
292                 return None
293             return (
294                 self._read_json(done_uri) or {}

```

```

295         ) # empty dict still terminates the poll (present == done)
296
297     def _read_json(self, uri: str) -> dict:
298         # Download to a real local dst FIRST: the cloud backends (gcs/s3) REQUIRE a dst
299         # raises ValueError ? unlike local's identity passthrough. Without an explicit
300         # would silently return {} for every gs://|s3:// object (the only backends this
301         try:
302             ref = parse_uri(uri)
303             with tempfile.TemporaryDirectory(prefix="cr-read-") as td:
304                 dst = os.path.join(td, ref.name or "obj.json")
305                 local = fetch(uri, dst, config=self._storage_config)
306                 with open(local, "r", encoding="utf-8") as f:
307                     data: Any = json.load(f)
308                 return data if isinstance(data, dict) else {}
309         except Exception:
310             LOG.warning("cloud-run: could not read %s", uri, exc_info=True)
311         return {}
312

```

14. user (2026-07-03T19:48:51.930Z)

```

1     """Abstract base for detector runners (TrackNet, WASB, future native, etc.).
2
3     This completes the Detector Runner Abstraction (low-regret item 6 continuation / item 1
4     of the post-observability re-prioritization) to de-risk the WSL/conda dependency for
5     the high-accuracy shuttle trajectory substrate used by tracknet_hybrid / wasb_hybrid.
6
7     The ABC lets us:
8     - Keep the current WSL adapters working unchanged for Windows users.
9     - Swap in (or A/B test) a Linux-native implementation later with zero changes to
10    the hybrid segmenters or the model-agnostic windowing brain.
11    - Provide a single place to evolve the healthcheck / error contract and thread
12    detector readiness into per-video reports.
13
14    Contract (deliberately matches what the concrete runners + hybrids + tests already did):
15    - run_predict(video_win_path: str, output_win_dir: str) -> Optional[str]
16      Run detector, return the Windows-side path to the produced trajectory CSV
17      (e.g. ".../clip_predict.csv" or ".../clip_wasb.csv"), or None on any failure.
18      The caller (hybrid) is responsible for output dir; impls may stage the video
19      into WSL fs for perf.
20
21    - healthcheck() -> tuple[bool, str]
22      Return (ok, message). Never raises. "ok" decides whether to proceed to run_predict.
23      Message is logged (and can be surfaced to reports or UX in future).
24      Intended to be cheap (env existence, import, GPU visibility, weights present).
25
26    Existing behavior is 100% preserved. The only additions are the inheritance and
27    the explicit common type for future implementers.
28
29    How to add a new runner (for docs / future contributors):
30    1. Subclass DetectorRunner in a new or existing module under pipeline/detectors/.
31    2. Implement run_predict and healthcheck (return (bool, str) tuple).
32    3. If it produces the same CSV schema (Frame,Visibility,X,Y), reuse or re-export
33       TrackNetRunner.parse_trajectory_csv and .trajectory_to_action_windows (they are
34       deliberately static and model-agnostic). If the new detector needs different
35       windowing, it can implement its own or we generalize later.
36    4. Lazy-import heavy native deps inside the methods (so importing the segmenter
37       registry never pulls torch/tf on a machine that doesn't have them).
38    5. Update the two hybrid segmenters (or introduce a detector= choice) to accept
39       DetectorRunner instances (they already do the right thing via duck-typing + the
40       type hints we add here).
41    6. Add unit tests that exercise the contract (isinstance, healthcheck shape, run

```

```

42     returns str|None) using mocks for the external bits ? see test_tracknet_runner.py.
43 7. No change to eval/calibration code paths (they consume the CSV + the static
44    windowing helpers directly).
45
46 Optional Linux-native path (the main de-risk goal):
47 A LinuxNativeRunner (or ONNXRunner) would:
48 - Skip all wsl / _stage / conda activate.
49 - Load weights locally (torch or onnx) or call a native binary.
50 - Write the exact same CSV format so trajectory_to_action_windows works verbatim.
51 - healthcheck can just check "weights file exists and torch importable + CUDA?".
52 This removes the WSL requirement for Linux users / future SaaS workers while
53 keeping the exact same candidate quality characteristics (or better, once native
54 weights are available).
55
56 Error surfacing into reports:
57 The healthcheck message is already logged by the hybrids on failure. The reporting
58 harvest sees the outcome as "0 candidates" + detector name. When we wire richer
59 segmenter Results (or pass detector_health into emit_indexer_report context), the
60 (ok, msg) can be recorded explicitly under segmentation.details or a new detector
61 stage. The ABC makes that future change local to one place.
62
63 See also:
64 - backend/pipeline/segmenters/{tracknet_hybrid,wasb_hybrid}.py (the two callers)
65 - tests/test_tracknet_runner.py (contract + pure logic tests; hybrids mock the runner)
66 - docs/TRACKNET_WSL_SETUP.md and CODE_MAP.md (2.2 Detectors section)
67 """
68
69 import logging
70 import os
71 import shlex
72 import subprocess
73 from abc import ABC, abstractmethod
74 from typing import Any, Callable, List, Optional, Tuple
75
76 logger = logging.getLogger(__name__)
77
78 # Frame extraction of a high-res clip can materialise ~15x the source size in PNGs
79 before the
80 # detector consumes them (STUDIO_MEDIA_LIFECYCLE.md D11). The WSL stage fs must hold
81 that peak.
82 WSL_FRAME_HEADROOM = 15.0
83
84 def wsl_tilde_quote(path: str) -> str:
85     """shlex.quote that still lets bash expand a leading `~/`.
86
87     ``shlex.quote("~/clips/x.mp4")`` single-quotes the whole path, so bash
88     treats ``~`` literally and ``cp``/``mkdir`` fail with "No such file or
89     directory" ? this broke live WSL staging for the default ``~/clips``
90     stage dir (found by the 2026-06-11 wasb_hybrid smoke run; the BACKLOG
91     "shlex-quote breaks ~ expansion" item). Quoting only the part after
92     ``~/`` keeps expansion AND protects spaces/specials in the remainder.
93     """
94     if path == "~":
95         return path
96     if path.startswith("~/"):
97         return "~/ " + shlex.quote(path[2:])
98     return shlex.quote(path)
99
100 class WslCommandMixin:
101     """Shared ``wsl -d <distro> bash -c 'source <conda_sh> && <cmd>'`` invocation.
102
103     Lives beside (not on) DetectorRunner: a future Linux-native runner must not
104     inherit WSL plumbing. Requires ``self.cfg`` with ``distro``, ``conda_sh``

```

```

105 and ``timeout_sec`` attributes (TrackNetConfig and WasbConfig both qualify).
106 ``timeout_sec == 0`` maps to None = wait forever (configs default 1800s).
107 """
108
109 cfg: Any
110
111 def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
112     full = f"source {self.cfg.conda_sh} && {inner_cmd}"
113     argv = ["wsl", "-d", self.cfg.distro, "bash", "-c", full]
114     logger.debug("WSL exec: %s", full)
115     return subprocess.run(
116         argv, capture_output=True, text=True, timeout=(self.cfg.timeout_sec or None)
117     )
118
119 # ---- cache hygiene helpers (shared by both WSL runners) -----
120 def _wsl_free_gb(self, wsl_path: str) -> Optional[float]:
121     """Best-effort free space (GB) on the WSL filesystem holding ``wsl_path``.
122
123     Returns None if it can't be determined (never raises). Used as a pre-run
124     disk guard so a long extraction doesn't fill the disk mid-flight.
125     """
126     try:
127         res = self._wsl_bash(
128             f"df -P -B1 {wsl_tilde_quote(wsl_path)} | awk 'NR==2{{print $4}}'"
129         )
130     except (subprocess.TimeoutExpired, OSError):
131         return None
132     try:
133         return int((res.stdout or "").strip()) / (1024**3)
134     except (ValueError, AttributeError):
135         return None
136
137 def _wsl_frame_stage_shortfall(
138     self, source_win_path: str, *, headroom: float = WSL_FRAME_HEADROOM
139 ) -> Optional[str]:
140     """Actionable error string if the WSL stage fs lacks ~`headroom`× the source
141     size for frame
142     extraction, else ``None`` (STUDIO_MEDIA_LIFECYCLE.md D11 ? BLOCK the local/WSL
143     frame stage).
144     BLOCKING replaces the old warn-only guard: extracting a 4K clip can balloon to
145     ~15× the
146     source in PNGs and fill the disk mid-run (a wasted long run that looks like a
147     quality fail).
148     Best-effort about MEASURING ? an unreadable source size or an undeterminable WSL
149     free-read
150     returns ``None`` so we never block on inability to measure ? but strict once a
151     real shortfall
152     is seen. ``min_free_gb`` (config) still acts as an absolute floor on top of the
153     headroom."""
154     try:
155         src_bytes = os.path.getsize(source_win_path)
156     except OSError:
157         return None
158     free_gb = self._wsl_free_gb(self.cfg.wsl_stage_dir)
159     if free_gb is None:
160         return None
161     src_gb = src_bytes / (1024**3)
162     need_gb = max(src_gb * headroom, float(getattr(self.cfg, "min_free_gb", 0.0) or
163     0.0))
164     if free_gb >= need_gb:
165         return None
166     return (
167         f"Low WSL disk at {self.cfg.wsl_stage_dir}: {free_gb:.1f} GB free, but frame
168         "

```

```

161         f"extraction needs ~{need_gb:.1f} GB (~{headroom:g}× the {src_gb:.2f} GB
source). "
162         f"Free space (e.g. python -m tools.cleanup_caches) and retry."
163     )
164
165     def _wsl_rm_rf(self, wsl_paths: List[str]) -> None:
166         """Best-effort recursive delete of WSL paths (post-success cleanup; never
raises)."""
167         paths = [p for p in wsl_paths if p]
168         if not paths:
169             return
170         quoted = " ".join(wsl_tilde_quote(p) for p in paths)
171         try:
172             self._wsl_bash(f"rm -rf {quoted}")
173         except (subprocess.TimeoutExpired, OSError) as e:
174             logger.warning("WSL cache cleanup failed (non-fatal): %s", e)
175
176     def wsl_source_unreadable_reason(self, src_mnt: str) -> Optional[str]:
177         """Actionable message if ``src_mnt`` is NOT readable from inside WSL, else None.
178
179         Google Drive (``G:``/Drive File Stream), UNC and many network/virtual
180         filesystems are invisible to WSL's ``/mnt`` ? staging a video from such a path
181         fails with a cryptic ``cp: cannot stat`` and the whole run silently produces no
182         trajectory (i.e. "0 rallies", a wasted run that looks like a quality result).
183         A cheap ``test -r`` preflight turns that into one clear, fixable error BEFORE
any
184         frame extraction. Best-effort: a flaky/timed-out probe returns None so the real
185         ``cp`` still gets its chance (we never block a genuinely-readable source).
186         """
187         try:
188             res = self._wsl_bash(f"test -r {shlex.quote(src_mnt)} && echo READABLE")
189         except (subprocess.TimeoutExpired, OSError):
190             return None
191         if "READABLE" in (res.stdout or ""):
192             return None
193         return (
194             f"video source is not readable from WSL at '{src_mnt}'. Google Drive, UNC, "
195             f"and other network/virtual filesystems are invisible to WSL's /mnt ? copy "
196             f"the video to a local disk (e.g. C:\\ or the WSL filesystem) and retry."
197         )
198
199
200     class DetectorRunner(ABC):
201         """Common interface for shuttle/ball trajectory detectors (WSL or otherwise)."""
202
203         @abstractmethod
204         def run_predict(
205             self, video_win_path: str, output_win_dir: str, video_id: Optional[str] = None
206         ) -> Optional[str]:
207             """Run inference and return path to trajectory CSV (or None on failure).
208
209             ``video_id`` (the file MD5) namespaces all staged/cached/output artifacts so two
210             videos that happen to share a filename cannot reuse each other's frames or CSV.
211             It is optional for back-compat; callers in the live pipeline always pass it.
212             """
213             pass
214
215         @abstractmethod
216         def healthcheck(self) -> Tuple[bool, str]:
217             """Quick check if the runner (env, weights, etc.) is usable.
218
219             Returns (ok, message). Never raises. Cheap to call before a long run.
220             """
221             pass
222

```

```

223 # De facto trajectory contract: the trajectory hybrids call these
224 # unconditionally after run_predict. Concrete runners reuse/re-export the
225 # model-agnostic TrackNetRunner staticmethods (same CSV schema + windowing);
226 # a runner with custom windowing supplies its own with the same shapes.
227 parse_trajectory_csv: Callable[..., Any]
228 trajectory_to_action_windows: Callable[..., Any]
229

```

15. assistant (2026-07-03T19:49:03.385Z)

I now have everything I need. The architecture is fully confirmed:

```

**Job lifecycle (the load-bearing fact):** `CloudRunCompute.run_infer` (cloud_run.py:179)
dispatches ONE Cloud Run **Job** execution per video via `run_job`. The Dockerfile ENTRYPOINT is
`python3 -m backend.worker` and the args are `infer --video-uri ... --out-uri ...`. So each
video = a fresh container = a fresh process that (a) pays lazy CUDA context init, (b) imports
torch 1.11 + Hydra + builds the WASB detector via `build_detector(cfg)` which loads the
`.pth.tar` weights from disk, (c) does cudnn autotune on the first forward. There is no
persistent process; `build_detector` is called exactly once per `run()` invocation
(wasb_infer.py:601), and `run()` is the whole process. Task #2 confirms "L4 Cloud Run **Job**"
(not Service).

```

```

**Grad/eval state:** `torch.no_grad()` IS entered (wasb_infer.py:619, wrapping the whole
detector loop). `model.eval()` is NOT visible in wasb_infer.py ? it is inside WASB's opaque
`build_detector(cfg)`. WASB's own eval config path (`configs/eval`) and the fact that this is
the upstream eval harness strongly implies the detector wrapper sets `.eval()` internally, but
wasb_infer.py does not assert it. `set_grad_enabled` is never called explicitly; `no_grad()`
covers it.

```

Let me do the honest cost accounting before returning the finding. The measured baseline is 727.9s for the whole Phase-2 with GPU <1% utilized. The warmup/JIT/context cost is a ONE-TIME fixed cost (a few seconds), whereas 727s is dominated by per-window feed-bound serialization across ~21k windows. So this lever is real but small relative to the headline number.

16. user (2026-07-03T19:49:42.195Z)

```

<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be
parsed as JSON.
You sent (first 200 of 6495 bytes): {"lever": "model-warmup-persist: (a) a dummy warmup forward
at healthcheck/startup to move lazy-CUDA-context + cudnn-autotune + first-JIT off the measured
path, and (b) recognizing the ephemeral Cloud
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters,
or truncated output. Retry with valid JSON.</tool_use_error>

```

17. assistant (2026-07-03T19:49:44.600Z)

I sent the fields wrapped in a `raw` string. Let me call the tool with the proper structured fields directly.

18. user (2026-07-03T19:50:22.104Z)

```

<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be
parsed as JSON.
You sent (first 200 of 6620 bytes): {"lever": "model-warmup-persist: (a) a dummy warmup forward
at healthcheck/startup so lazy CUDA-context + cuDNN autotune + first-forward JIT/kernel-cache
land OUTSIDE the measured detection timer; (b)
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters,
or truncated output. Retry with valid JSON.</tool_use_error>

```

19. assistant (2026-07-03T19:50:25.605Z)

Let me pass the fields as proper individual JSON fields, keeping the text concise to avoid escaping issues.

20. user (2026-07-03T19:50:55.566Z)

<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be parsed as JSON.
You sent (first 200 of 5649 bytes): {"lever": "model-warmup-persist: (a) a warmup forward at startup so lazy CUDA-context init + cuDNN autotune + first-forward JIT/kernel-cache land OUTSIDE the measured detection timer; (b) recognizing
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters, or truncated output. Retry with valid JSON.</tool_use_error>

21. assistant (2026-07-03T19:51:00.085Z)

The harness is showing me a `raw` wrapper, which suggests my tool input is being double-wrapped. Let me call the tool passing each parameter as a native argument rather than a serialized JSON string, keeping content tight.

22. user (2026-07-03T19:51:31.991Z)

<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be parsed as JSON.
You sent (first 200 of 5997 bytes): {"lever": "model-warmup-persist: (a) a warmup forward at startup so lazy CUDA-context init + cuDNN autotune + first-forward JIT/kernel-cache land OUTSIDE the measured detection timer; (b) recognizing
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters, or truncated output. Retry with valid JSON.</tool_use_error>

23. assistant (2026-07-03T19:51:35.375Z)

The backtick characters in my content (`` `python3 ...` ``) may be the problem, or the harness keeps interpreting my input as a `raw`-wrapped string. The tool expects native structured arguments. Let me provide them directly as proper JSON object fields with no backticks and careful escaping.

24. user (2026-07-03T19:52:13.889Z)

<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be parsed as JSON.
You sent (first 200 of 7009 bytes): {"lever": "model-warmup-persist: (a) a warmup forward at startup so lazy CUDA-context init + cuDNN autotune + first-forward kernel-cache land OUTSIDE the measured detection timer; (b) recognizing the
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters, or truncated output. Retry with valid JSON.</tool_use_error>